

Robust analog two-qubit gates from deep reinforcement learning: an independent reproduction and family-resolved assessment

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(Dated: October 2, 2026)

Reinforcement learning has been proposed as a unified route to analog quantum gate design, in which accuracy, leakage suppression, runtime and robustness against stochastic control errors are optimized jointly. We report an independent, from-scratch reimplementation of the Universal cost Function control Optimization (UFO) framework of Niu *et al.* [npj Quantum Inf. **5**, 33 (2019)] for the two-qubit gate family $N(\alpha, \alpha, \gamma) = \exp[i(\alpha\sigma^x\sigma^x + \alpha\sigma^y\sigma^y + \gamma\sigma^z\sigma^z)]$ on a three-level gmon model, and compare its trust-region policy optimization (TRPO) agent with a direct gradient (GRAPE-style, Adam) optimizer that uses the same Hamiltonian, the same objective and the same training noise. The framework produces high-fidelity analog gates, but three of its central claims are not supported in our implementation, and for two of them the reason does not depend on the implementation. Against per-step white control noise, the open-loop robustness of any pulse is fixed to leading order by its nominal fidelity and its duration — the white-noise limit of the filter-function picture — at a rate that we compute from the Hamiltonian and verify on every pulse: no training procedure can buy robustness through the pulse shape. A direct optimizer confirms this over eight seeds and three durations. Trained under white noise it gains nothing in 48 runs out of 48, with the leakage bound evaluated on the noisy or on the noise-free controls; trained under quasi-static noise, where the shape does enter, it lowers the sensitivity of its pulses by about a fifth. Two TRPO agents that differ only in the training noise agree: their deployed pulses are equally robust within the seed-to-seed spread, and what stochastic training changes is the policy’s reaction to noisy states, which an open-loop pulse cannot use. Three further pairs of agents, retrained with the leakage bound taken from the noise-free controls, give the same result: the noise-trained pulses are the more robust ones in about a quarter of the matched iterations, by the margin that their durations and nominal fidelities predict. In our implementation a single-start direct optimizer also beats the policy-gradient controller on every nominal metric and at every noise strength, which is a statement about our agents and not about reinforcement learning in general. Finally, every member of the benchmark subfamily $N(\alpha, \alpha, \pi/2)$ is an exchange rotation by $\alpha - \pi/2$, so the order-of-magnitude runtime gain reported near $\alpha = \pi/2$ sits around the identity. Against a target-specific two-CNOT synthesis the learned pulses are at most twice as fast, and the direct optimizer up to 3.75 times, at nominal fidelities of 0.98 or more; its pulses sit at the limit set by the model’s control bandwidth, which the gate times of the synthesis reference do not respect. All code and data are public.

I. INTRODUCTION

A quantum algorithm is written as a sequence of logical unitaries, but a quantum processor executes continuous-time dynamics generated by an experimentally accessible Hamiltonian. On pre-fault-tolerant superconducting hardware the translation between the two is not a formality: coherence budgets, calibration drift, waveform distortion and multilevel structure all limit which circuits can actually be run [1–3]. Quantum control is therefore not a fine-tuning layer on top of compilation, but one of the binding constraints on usable computational depth.

This observation motivates a shift of perspective. Rather than decomposing a target two-qubit unitary into a fixed universal gate alphabet and paying the runtime cost of every elementary operation, one may attempt to realize it directly through a single shaped analog pulse that exploits the native interactions of the device [4, 5]. The appeal is concrete: if the target is well aligned with

the hardware’s own entangling generator, the analog implementation can be substantially shorter than any compiled circuit. The difficulty is equally concrete. Superconducting modes are weakly anharmonic, so fast or aggressively modulated pulses drive population out of the computational subspace [6, 7]; and the pulse that reaches the device is never the pulse that was designed, because amplitudes fluctuate and Hamiltonian parameters are known only approximately.

Niu *et al.* [4] proposed to attack these requirements simultaneously rather than in sequence. Their Universal cost Function control Optimization (UFO) framework combines three ingredients: an analytic leakage estimate derived from a time-dependent Schrieffer–Wolff transformation (TSWT), a scalar objective that adds leakage, boundary and runtime penalties to gate infidelity, and a continuous-action reinforcement-learning (RL) formulation trained with trust-region policy optimization (TRPO) [8] in an environment where control noise is injected during training. The reported outcome was strong: up to one order of magnitude lower average gate infidelity than an RL agent trained without noise, around two orders of magnitude lower than a stochastic gradient-

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descent (SGD) baseline, a fidelity standard deviation two orders of magnitude smaller than either, and a one-order-of-magnitude runtime reduction relative to optimal gate synthesis on the subfamily $N(\alpha, \alpha, \pi/2)$ for $\alpha \in [1.2, 1.7]$. The authors left one question explicitly open: whether the advantage of RL persists when gradients are cheap enough for gradient-based optimization to apply.

These are consequential claims, and they have been influential in establishing RL as a credible language for robust pulse design [9–11]. They have not, to our knowledge, been subjected to an independent reimplementa-tion. That is the purpose of this work.

What this paper does. We rebuild the framework from scratch — multilevel gmon simulator, second-order TSWT leakage estimator, exact bandwidth filter, UFO objective, TRPO agent and stochastic environment — and use it to ask four questions. (i) Does the frame-work reproduce under an independent implementation, and with what quantitative fidelity? (ii) How does the policy-gradient controller compare with a direct pulse op-timizer that computes exact gradients through the same simulator, with the same objective and the same train-ing noise — the question Ref. [4] left open? (iii) Does stochastic training make the deployed pulses more ro-bust, and if not, why not? (iv) What does the runtime comparison with gate synthesis actually measure, given the structure of the benchmark family? Along the way we test how much the conclusions depend on the free parameters the original paper does not pin down, the actor–critic architecture and the UFO weights.

What we find. The framework works as a pulse gen-erator, but its central claims are not supported by con-trolled tests in our implementation. Items 3 and 4 below do not depend on the implementation; items 1 and 2 do.

1. On the representative target $N(2.2, 2.2, \pi/2)$, a single-start Adam optimizer beats both TRPO agents on *every* nominal metric — against the nominal agent, total cost 0.158 versus 0.298, run-time 70 versus 116 ns and leakage 2.8×10^{-4} ver-sus 1.6×10^{-3} — and one of its two plans is the most robust controller at every noise strength. This compares one gradient optimizer with our agents, whose curriculum fails on a third of the family (item 4); better-trained agents could narrow the gap.
2. Stochastic training does not make the deployed pulse more robust. Two TRPO agents that share seed, architecture, objective and budget and dif-fer only in the training noise produce open-loop pulses of the same robustness: over eighty matched iterations the noise-trained pulse is the more ro-bust one in only 19–25% of them, at every noise strength. The difference is of the size of the seed-to-seed spread, so its sign is not established; what the data exclude is a gain of the reported size. Our noise-trained agent also saw a leakage bound that responds to the noise (Sec. VII C). Three further pairs of agents, with three seeds and the bound

taken from the noise-free controls, repeat the re-sult: the noise-trained pulse is the more robust one in 24–28% of the matched iterations at 1 MHz.

3. The reason is structural. Against per-step white control noise, the open-loop robustness of any pulse is fixed to leading order by its nominal fidelity and duration, Eq. (21), at a rate that we com-pute from the Hamiltonian and verify on every pulse we produced. In addition, in our implemen-tation the noisy objective is dominated by the leak-age bound’s response to the noise, which acts as a runtime penalty, and what stochastic training does change is the policy’s reaction to noisy states: in closed loop the noise-trained agent is the more ro-bust one, but a deployed pulse cannot use feed-back. The same test with the direct optimizer, over eight seeds and three durations, agrees: white-noise training never helps, whether or not the leakage bound sees the noise, and training against quasi-static noise, where the pulse shape does enter, low-ers the sensitivity by about a fifth (Sec. V E).
4. Every member of the subfamily $N(\alpha, \alpha, \pi/2)$ is an exchange rotation by $\alpha - \pi/2$. The target at $\alpha = \pi/2$ is the identity, the order-of-magnitude runtime gain of Ref. [4] lies around it, and every non-local member needs two CNOTs, not three. Against that target-specific synthesis the converged curriculum pulses are 1.0–2.0 $\times$ faster; the curriculum fails to converge on a third of the family, and an exact mirror symmetry of the problem shows that these failures are the optimizer’s, not the targets’.
5. The runtime advantage over compiled synthesis be-longs to analog pulse-level control, not to the learn-ing method: the direct optimizer alone is faster at every non-trivial target (1.9–3.75 $\times$ over two-CNOT synthesis, also for its 21 solutions with fi-delity above 0.98), and its pulse length tracks the exchange angle $|\alpha - \pi/2|$, close to the shortest time in which the bandwidth-limited coupling can de-liver it. Under the same bandwidth limit a single-qubit π rotation takes at least 54 ns, not the 20 ns assumed by the synthesis reference.

Section II states the framework; Sec. III gives the implementation and the complete parameter set; Ap-pendix D fixes the free parameters through two sweeps; Secs. IV–VI present the nominal, robustness and runtime results; Sec. VII confronts them with the original claims.

II. CONTROL FRAMEWORK

A. Physical model and control channels

The platform is the gmon architecture, sketched in Fig. 1: two weakly anharmonic superconducting modes

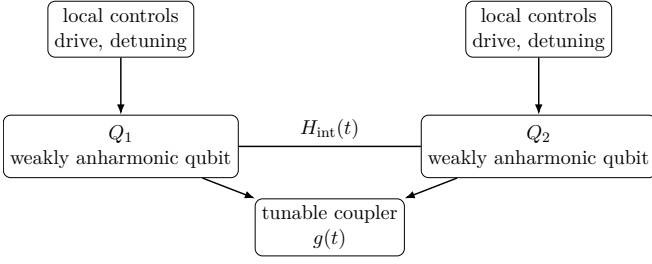


FIG. 1. The gmon control setting. Two weakly anharmonic modes are driven locally and coupled through a tunable interaction $g(t)$. Local addressability together with a time-dependent coupling gives the seven control channels of Eq. (2).

with local drive and frequency control, coupled through a flux-tunable interaction [5]. The tunability of the coupling is what makes the platform interesting for analog gate design, because the entangling interaction becomes a control resource rather than a fixed background.

After circuit quantization, transformation to the interaction picture and a rotating-wave approximation, the effective two-mode Hamiltonian is [4]

$$\begin{aligned} \hat{H}_{\text{RWA}}(t) = & \frac{\eta}{2} \sum_{j=1}^2 \hat{n}_j (\hat{n}_j - 1) + g(t) (\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2) \\ & + \sum_{j=1}^2 \delta_j(t) \hat{n}_j \\ & + \sum_{j=1}^2 i f_j(t) \left(\hat{a}_j e^{i\phi_j} - \hat{a}_j^\dagger e^{-i\phi_j} \right), \end{aligned} \quad (1)$$

with $\hat{n}_j = \hat{a}_j^\dagger \hat{a}_j$. The four terms carry distinct physical roles. The anharmonicity η breaks the harmonic ladder and makes the lowest two levels of each mode a usable qubit; because η is finite, the system stays intrinsically multilevel and leakage remains a real error channel. The exchange term is the entangling resource. The detunings act diagonally and shape relative phase accumulation. The drives generate local rotations. The available controls are therefore

$$\mathbf{u}(t) = \{g(t), \delta_1(t), \delta_2(t), f_1(t), f_2(t), \phi_1(t), \phi_2(t)\}. \quad (2)$$

Projected onto the logical subspace, Eq. (1) becomes

$$\begin{aligned} \hat{H}_{\text{qb}}(t) = & \frac{g(t)}{2} (\sigma_1^x \sigma_2^x + \sigma_1^y \sigma_2^y) \\ & + \sum_j \left[\frac{\delta_j}{2} \sigma_j^z - f_j (\sin \phi_j \sigma_j^x + \cos \phi_j \sigma_j^y) \right], \end{aligned} \quad (3)$$

which shows that the coupling generates an XY -type entangler while the local channels supply arbitrary single-qubit rotations — enough for universal two-qubit control. Equation (3) is an interpretive tool only; all dynamics in this work are propagated with the full multilevel Eq. (1), because the whole point is that leakage must not be projected away.

B. Subspaces and the TSWT leakage bound

In the Fock basis $|n_1 n_2\rangle$ the computational sector is $\Omega_0 = \text{span}\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ and the leading leakage sector is $\Omega_1 = \text{span}\{|02\rangle, |20\rangle, |12\rangle, |21\rangle\}$. Writing the physical space as $\mathcal{H} = \bigoplus_\alpha \Omega_\alpha$ and decomposing

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) + \hat{H}_2(t), \quad (4)$$

the diagonal drift $\hat{H}_0$ sets the subspace energies, $\hat{H}_1$ is block-diagonal (evolution within each sector) and $\hat{H}_2$ is block-off-diagonal (direct leakage). With Δ the minimum gap between Ω_0 and the leakage sectors and $\epsilon \sim \|\hat{H}_1\| \sim \|\hat{H}_2\|$ the control scale, the regime of interest is $\epsilon/\Delta \ll 1$.

The essential difficulty is that $\hat{H}_2$ cannot simply be switched off: the same controls that produce the desired logical rotations also generate the inter-subspace coupling. Leakage suppression is therefore a control problem, not a projection argument. The TSWT addresses it by moving to a rotated frame $|\tilde{\psi}\rangle = e^{-\hat{S}(t)} |\psi\rangle$ with anti-Hermitian $\hat{S}$, in which

$$\hat{\mathbb{H}}(t) = -i \frac{d e^{-\hat{S}}}{dt} e^{\hat{S}} + e^{-\hat{S}} \hat{H} e^{\hat{S}}. \quad (5)$$

The first term is the price of a moving basis: in a driven system, leakage is generated not only by direct coupling but also by the non-adiabatic motion of the frame itself. Expanding $\hat{S} = \epsilon \hat{S}_1 + \epsilon^2 \hat{S}_2 + \dots$ and cancelling block-off-diagonal terms order by order leaves, at second order, a residual $\hat{H}_{\text{od}} = O(\epsilon^3/\Delta^2)$, from which the accumulated coherent leakage over a gate of duration T is bounded by

$$\begin{aligned} L_{\text{tot}}(T) \lesssim & \frac{\|\hat{H}_{\text{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\text{od}}(T)\|}{\Delta(T)} \\ & + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\text{od}}(t)}{dt^2} \right\| dt. \end{aligned} \quad (6)$$

A derivation is summarized in Appendix A. The structure of Eq. (6) is what makes it useful for optimization rather than merely for diagnosis. The two boundary terms penalize residual off-diagonal coupling at the pulse endpoints, and therefore push the controls toward configurations where the dressed and bare bases coincide at $t = 0$ and $t = T$. The integral term penalizes rapid modulation of $\hat{H}_{\text{od}}$ and is the quantitative expression of the speed-leakage trade-off: a pulse can be small in amplitude at every instant and still leak, if it changes too quickly. Crucially, Eq. (6) is written in terms of the Hamiltonian and its derivatives, so it can be accumulated online during propagation instead of requiring a separate population measurement at the end of the pulse.

C. The UFO objective

Let $U(T) = \mathcal{T} \exp[-i \int_0^T \hat{H}_{\text{RWA}}(t) dt]$ and let U_{proj} be its restriction to Ω_0 . The gate fidelity with respect to

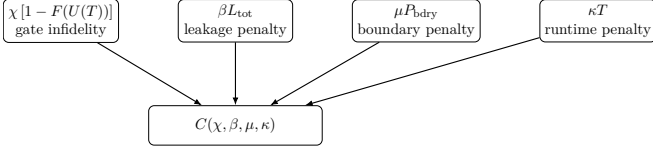


FIG. 2. The UFO objective, Eq. (8), scalarizes four competing physical requirements: logical accuracy, accumulated leakage, boundary regularity and total runtime.

U_{target} is

$$F = \frac{1}{16} |\text{Tr}(U_{\text{proj}}^\dagger(T) U_{\text{target}})|^2, \quad (7)$$

which equals 1 exactly when the implemented gate matches the target up to a global phase. The full objective adds three penalties to this term, as summarized in Fig. 2 [4]:

$$C(\chi, \beta, \mu, \kappa) = \chi[1 - F] + \beta L_{\text{tot}} + \kappa T + \mu \sum_{t \in \{0, T\}} [g(t)^2 + f_1(t)^2 + f_2(t)^2]. \quad (8)$$

The boundary term enforces vanishing envelopes at the pulse edges, which both permits clean gate concatenation and reduces errors from deviations of the rotating-wave approximation; the linear runtime penalty encodes the fact that long gates buy more exposure to decoherence and drift.

The design choice that matters here is that all of these are *soft* constraints. Much earlier leakage-mitigation work proceeded by disabling control channels or imposing structural restrictions on the admissible pulse shape, which suppresses leakage at the price of controllability. UFO keeps every channel available and penalizes undesirable behaviour instead, leaving the optimizer free to trade the four objectives against one another. This also means the framework is modular: on a different platform the individual terms can be replaced without changing the logic.

D. Reinforcement-learning formulation

The pulse is built sequentially. Time is discretized in steps Δt ; the state presented to the agent is a representation of the propagator accumulated so far, $s_n \sim U_n$; the action is the control vector for the next interval, $a_n = (g_n, \delta_{1,n}, \delta_{2,n}, f_{1,n}, f_{2,n}, \phi_{1,n}, \phi_{2,n})$; and the environment propagates

$$U_{n+1} = \exp[-i\Delta t(\hat{H}_{n+1} + \delta\hat{H}_{n+1})]U_n, \quad (9)$$

where $\delta\hat{H}$ is the stochastic perturbation used in noisy training. The reward is the negative running UFO cost, $r_n = -c_n$, so maximizing expected return minimizes the expected objective. An episode terminates at

$\min(T_{\text{max}}, T_{\text{stop}})$, where T_{stop} is the first time the cost falls below a threshold; the agent thus co-optimizes pulse shape *and* stopping time, which is a structural difference from fixed-horizon pulse optimization.

Using the accumulated unitary as the state is a deliberate choice: the agent reasons in terms of how much of the target operation has already been built, rather than in terms of local control variables, which matches the structure of gate synthesis.

Optimization uses TRPO [8], which maximizes the importance-weighted surrogate

$$\mathcal{L}(\theta) = \mathbb{E}_{\pi_{\theta_{\text{old}}}} \left[\frac{\pi_{\theta}(a|s)}{\pi_{\theta_{\text{old}}}(a|s)} \hat{A}(s, a) \right] \quad \text{s.t.} \quad \mathbb{E}_s[D_{\text{KL}}] \leq \delta_{\text{KL}}. \quad (10)$$

The trust-region constraint matters here for a specific reason: the UFO reward couples four competing terms and the environment is stochastic, so the landscape is irregular and a single over-large gradient step can destroy previously learned behaviour. Constraining the step in distribution space buys stability at the cost of speed.

Robustness enters through the environment rather than through an outer loop. In noise-optimized training, Gaussian fluctuations are added to the control amplitudes at every step, so the objective becomes an expectation over both the policy and the noise, $J(\theta) = -\mathbb{E}_{\tau, \xi}[C]$. The intended effect is that pulse trajectories fragile under small perturbations earn poor average return and are disfavoured, so that stochastic training acts as an implicit regularizer. Throughout this paper we call *nominal* an agent trained in the deterministic environment and *noise-optimized* one trained in the stochastic environment. They differ only in the training distribution — same architecture, same objective, same algorithm — which is what makes their comparison informative.

Two details of this construction matter later. First, the agent observes the propagator produced by the *noisy* dynamics, so during training it can react to the realized noise; the pulse that is deployed and evaluated is the open-loop plan of its mean action in the noise-free environment, which cannot. Second, every term of Eq. (8) is evaluated on the noisy trajectory, including the leakage bound, whose time derivatives respond to the noise itself (Sec. V D).

E. Target family and synthesis reference

Benchmarking uses the family

$$N(\alpha, \alpha, \gamma) = \exp[i(\alpha\sigma_1^x\sigma_2^x + \alpha\sigma_1^y\sigma_2^y + \gamma\sigma_1^z\sigma_2^z)]. \quad (11)$$

The equal XX and YY coefficients mirror the exchange structure of the gmon coupling, so the family is deliberately close to the hardware's native interaction while remaining rich: up to single-qubit rotations it contains SWAP, iSWAP, CNOT and CZ, as well as the fermionic SWAP and Givens rotations that appear in Jordan–Wigner-mapped fermionic simulation [4]. Because it is

continuously parametrized, one can study an entire manifold of related targets rather than isolated cases.

The runtime reference of Ref. [4] is a generic circuit decomposition: three CNOTs and four layers of single-qubit rotations, the Vatan–Williams construction, which is optimal for an *arbitrary* two-qubit unitary [12]. With the gate times assumed there, $T_{1q} = 20$ ns and $T_{\text{CNOT}} = 45$ ns, it takes

$$T_{\text{syn}} \approx 215 \text{ ns.} \quad (12)$$

We keep this number for comparability, but it is a generic reference, not the optimal synthesis of each member of the family: for the subfamily studied in Sec. VI two CNOTs suffice, and three members need no entangling gate at all (Sec. II F).

F. Structure of the subfamily $\gamma = \pi/2$

The runtime study uses $\gamma = \pi/2$, and there the two-qubit structure can be stated exactly. Since $\exp(i\frac{\pi}{2}\sigma_1^z\sigma_2^z) = i\sigma_1^z\sigma_2^z$ and $\exp[i\frac{\pi}{2}(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y)] = \sigma_1^z\sigma_2^z$, every member is, up to a global phase, a pure exchange rotation,

$$N(\alpha, \alpha, \pi/2) = i \exp[i\varepsilon(\sigma_1^x\sigma_2^x + \sigma_1^y\sigma_2^y)], \quad \varepsilon = \alpha - \frac{\pi}{2}, \quad (13)$$

which is the decomposition noted in Ref. [4] with its local Z rotations absorbed. Four consequences are used below.

(i) *Local members.* At $\alpha = \pi/2$ the target is the identity, and at $\alpha = 0$ and $\alpha = \pi$ it is the local gate $\sigma_1^z\sigma_2^z$. Because the fidelity (7) ignores global phases, a pulse that does nothing implements the $\alpha = \pi/2$ target exactly.

(ii) *Entangling class and CNOT count.* In the Weyl chamber [13] each member lies at $(a, a, 0)$ with $a = \min(|\varepsilon|, \pi/2 - |\varepsilon|)$, so the most entangling members, $|\varepsilon| = \pi/4$, are iSWAP-equivalent. Because one canonical coordinate vanishes, every non-local member needs two CNOTs rather than three [14, 15]. With the gate times above and three single-qubit layers, the target-specific synthesis time is

$$T_{\text{syn}}^{\text{tgt}}(\alpha) = \begin{cases} 0 & \alpha = \pi/2, \\ 20 \text{ ns} & \alpha \in \{0, \pi\}, \\ 150 \text{ ns} & \text{otherwise.} \end{cases} \quad (14)$$

On processors whose native entangler is itself a tunable exchange gate, as in the fSim gate set of gmon-type devices [16], each member is a single native two-qubit operation, and the synthesis overhead disappears altogether.

(iii) *Mirror symmetry.* The parity of the first mode, $P_1 = (-1)^{\hat{n}_1}$, maps $\hat{a}_1 \rightarrow -\hat{a}_1$. Conjugating Eq. (1) by P_1 therefore sends $g \rightarrow -g$ and $\phi_1 \rightarrow \phi_1 + \pi$ and leaves everything else unchanged, including the leakage structure and the boundary penalty, while on the qubits it maps $\varepsilon \rightarrow -\varepsilon$. The control problems at α and $\pi - \alpha$ are thus exactly equivalent: a pulse for one, with those

two substitutions, is a pulse for the other with the same cost.

(iv) *Required exchange area.* In the qubit projection (3), the coupling alone generates Eq. (13) once $|\int g dt| = 2|\varepsilon|$, with g in angular units. At the 20 MHz bound this takes at most 25 ns for any member. Pulse durations are therefore set by the bandwidth filter, the boundary conditions and the cost trade-off rather than by the amount of entangling action required: each pole of the filter (15) has a time constant $1/(\pi B) \approx 32$ ns, and a filtered step needs about 52 ns to reach half its amplitude. Section VIE makes this quantitative: the largest exchange area that a bounded control can deliver through the filter in a given time is a linear program, and it sets the scale of every runtime in this paper.

III. NUMERICAL IMPLEMENTATION

A. Simulator

Each mode is truncated to three levels, giving $d_{\text{phys}} = 3^2 = 9$ and $d_{\text{comp}} = 4$. Hamiltonians are kept in rad ns^{-1} internally; one-step propagation uses exact diagonalization of the Hermitian generator rather than a series expansion. The base anharmonicity is $\eta = -200$ MHz, which also sets the gap scale entering Eq. (6). Control amplitudes are bounded at 20 MHz, well below $|\eta|$ so that the perturbative regime $\epsilon/\Delta \ll 1$ is respected, and phases are mapped to $[0, 2\pi]$.

Before reaching the Hamiltonian, the piecewise-constant controls pass through the two-pole double-exponential filter of Ref. [4], implemented exactly: with $a = \exp(-\pi B/f_s)$, $B = 10$ MHz the bandwidth and $f_s = 1/\Delta t$ the sampling rate,

$$\tilde{u}_k = (1 - a)^2 u_k + 2a \tilde{u}_{k-1} - a^2 \tilde{u}_{k-2}. \quad (15)$$

This is not cosmetic. It ensures that the optimizer cannot exploit arbitrarily sharp control transitions that no waveform generator could produce, and it interacts directly with the non-adiabatic term of the leakage bound.

The leakage estimator implements the second-order TSWT numerically: at each step it decomposes H_t into H_0, H_1, H_2 , solves for the generators S_1 and S_2 on the off-diagonal blocks, estimates $\dot{S}_1, \dot{S}_2$ by finite differences, forms the residual $H_{\text{od}}^{(3)} = [H_1, S_2] + \frac{1}{3}[[H_2, S_1], S_1] - i\dot{S}_2$, and accumulates the boundary and integral contributions of Eq. (6) incrementally (Appendix B). The incremental form matters for cost: a naive implementation rescans the trajectory history at every step and is $O(T^2)$ per rollout.

B. Noise model

At each step the realized control is $\mathbf{u}^{(\text{noisy})}(t_k) = \mathbf{u}(t_k) + \boldsymbol{\xi}(t_k)$ with $\boldsymbol{\xi}$ drawn independently from a zero-

mean Gaussian of standard deviation σ_{noise} . Perturbations are applied to the coupling g , the detunings $\delta_{1,2}$ and the drive amplitudes $f_{1,2}$, and additionally to the anharmonicity η ; phases are left noise-free. Including η is a deliberate feature of the model: it probes sensitivity not only to imperfect pulse delivery but also to uncertainty in a device parameter that the controller cannot actuate at all. The noise is added after the bandwidth filter and is independent from one 2 ns step to the next, so it is white up to the 250 MHz Nyquist frequency of the grid, above the 200 MHz gap to the leakage sector. Ref. [4] discretizes its pulses more finely, in “around one thousand time steps”; the filter time constant of 32 ns is well resolved at 2 ns, so the noise-free control problem is essentially the same, but the step changes the meaning of σ_{noise} (Sec. VII C).

Training noise and evaluation noise are kept separate. The nominal TRPO agent is optimized without perturbations. The noise-optimized TRPO agent and the Adam baseline are both optimized under this perturbation model at $\sigma_{\text{train}} = 1$ MHz (Sec. III C). All controllers are then probed *a posteriori* on the evaluation grid of Sec. V.

C. Controllers compared

Direct gradient baseline (Adam). For a fixed target and a fixed horizon T , the full discretized trajectory $\{g_\ell, \delta_{j,\ell}, f_{j,\ell}, \phi_{j,\ell}\}_{\ell=0}^{N_T-1}$ is optimized directly with Adam [17] on the same multilevel model, the same filter and the same objective Eq. (8), differentiating exactly through the simulation — a GRAPE-style direct pulse search [3, 18] rather than a literal reproduction of the canonical algorithm. It is trained under the same noise model as the noise-optimized agent: at every Adam step the loss is the cost of one trajectory perturbed at $\sigma_{\text{train}} = 1$ MHz, a single-sample stochastic gradient of $\mathbb{E}_\xi[C]$. Since the horizon is not a free variable of a single run, the procedure is repeated over $T \in \{40, 50, \dots, 180\}$ ns, with 400 Adam iterations per horizon, learning rate 0.03 and a single deterministic initialization (all amplitudes zero, phases π) — 3.3×10^5 simulator steps in all, against 9.5×10^7 for each TRPO agent, a factor of about 290. Within a horizon the stored plan is the iterate with the lowest sampled loss, and the optimizer’s own choice of horizon is the one with the lowest such loss. Because single-sample losses are noisy, we also re-evaluate every horizon’s plan in the deterministic simulator and report both selections. One horizon takes about 100 s on one CPU core.

By the standards of Ref. [4] this baseline is modest. The SGD baseline there also used Adam and a noisy objective, but averaged ten noisy trajectories per gradient step and received about 2000 random restarts per target. What differs here is that the gradients are exact and are computed through the same filtered multilevel simulator in which TRPO is trained. The comparison therefore

addresses the question Ref. [4] left open — whether the advantage of reinforcement learning persists when gradient estimation is cheap enough for gradient-based optimization to apply — rather than whether a weak baseline can be beaten. The baseline is a family of fixed-horizon problems, whereas TRPO learns a variable-time policy; this structural difference is part of what the comparison tests.

Nominal and noise-optimized TRPO. Both use the same actor-critic pair. The observation vectorizes the current 9×9 propagator into real and imaginary parts and appends the normalized time index,

$$s_t = (\Re \text{vec}(U_t), \Im \text{vec}(U_t), t/T_{\text{max}}), \quad d_{\text{obs}} = 163. \quad (16)$$

Actor and critic are multilayer perceptrons with hidden sizes (64, 32, 32) and tanh activations, as in Ref. [4]. The actor emits a mean $\mu_\theta(s_t) \in \mathbb{R}^7$ with a state-independent trainable $\log \sigma$; raw actions $\tilde{a}_t = \mu_\theta + \sigma \odot \zeta_t$, with ζ_t standard normal, are squashed by tanh into $[-1, 1]^7$ and rescaled to physical bounds. The critic outputs a scalar baseline for advantage estimation. At every iteration the mean-action rollout in the noise-free environment is stored as a control plan; the controller reported for an agent is its plan with the lowest nominal cost over the whole training run.

Gate synthesis. The generic compiled reference of Eq. (12) and the target-specific one of Eq. (14), used for runtime only.

D. Metrics

Nominal quality is reported through the gate fidelity Eq. (7), the TSWT leakage estimate L_{tot} with its boundary/integral split, the boundary penalty, the effective runtime and the total cost Eq. (8).

Robustness is characterized by two distinct quantities, and the distinction is worth stating explicitly because they are often conflated. The first is the average fidelity of the *noise-averaged channel* $\mathcal{E}$, estimated from the Monte Carlo ensemble of noisy projected unitaries $\{K_j\}$ through the standard formula [19]

$$\bar{F} = \frac{\sum_j \text{Tr}(U_{\text{target}} U_j^\dagger U_{\text{target}}^\dagger \mathcal{E}(U_j)) + d^2}{d^2(d+1)}, \quad d = 4, \quad (17)$$

with $\{U_j\}$ a two-qubit Pauli basis. The second is the variance across realizations of the per-shot gate fidelity,

$$\text{Var}(F) = \mathbb{E}_\xi[(F_\xi - \mathbb{E}_\xi[F_\xi])^2]. \quad (18)$$

A controller with good mean fidelity but large variance is experimentally fragile, so neither quantity alone settles the question.

E. Parameter set

Table I collects every setting used. We report per-experiment values rather than a single idealized configuration, because the runs differ in budget and stopping cost, and the two single-target agents were trained in two stages; conflating them would misstate what was actually done.

F. Free parameters

Two ingredients of the framework are not determined by the physics: the capacity of the actor-critic networks and the relative weights in Eq. (8). Both could silently drive the comparisons that follow, so we fixed them first, with two sweeps on the representative target $N(2.2, 2.2, \pi/2)$ under a reduced budget (Appendix D). Neither sweep gives a reason to depart from the published choices: the layout (64, 32, 32) and the weights

$$(\chi, \beta, \mu, \kappa) = (10, 10, 0.2, 0.1) \quad (19)$$

are within the seed-to-seed spread of the best configurations we found, and we use them throughout. The sweeps also give that spread: three seeds of the same configuration reach $\bar{F} = 0.959, 0.962$ and 0.965 at $\sigma_{\text{noise}} = 1$ MHz. Differences of a few 10^{-3} in $\bar{F}$ between two individual runs are therefore not significant on their own, which sets the resolution of every single-seed comparison below.

IV. SINGLE-TARGET BENCHMARK

All controllers are first compared on

$$U_{\text{target}} = N(2.2, 2.2, \pi/2), \quad (20)$$

a target that is neither near-identity nor a degenerate special case, that sits in the family used for the original robustness analysis, and that isolates optimizer differences by holding target difficulty fixed.

A. Direct gradient baseline

The Adam baseline is trained under noise, but its plans are compared here in the deterministic simulator. Under that re-evaluation the best solution is at $T^* = 70$ ns, with nominal cost 0.1576, fidelity 0.99662, total leakage 2.82×10^{-4} and boundary penalty 4.28×10^{-3} . Direct pulse optimization thus reaches fidelity comfortably above 99% with leakage in the 10^{-4} regime — exactly the scale the leakage-aware objective is designed to produce. The optimizer’s own noisy objective selects 60 ns instead, a plan with fidelity 0.98205; Sec. VD explains why it prefers the shorter horizon. The baseline is trained under noise to match the noise-optimized agent; optimized without noise at the same 70 ns it does better

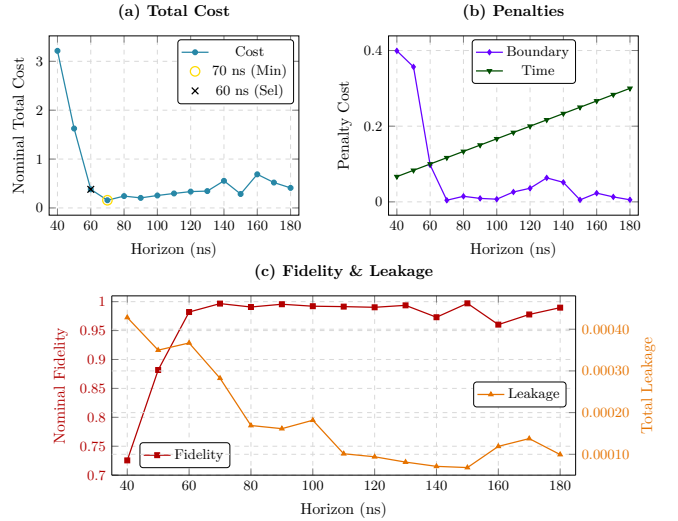


FIG. 3. Adam baseline on $N(2.2, 2.2, \pi/2)$ versus gate horizon. (a) Nominal total cost, with the optimum at 70 ns and the plan selected by the noisy objective at 60 ns both marked. (b) Boundary and runtime penalties. (c) Nominal fidelity and total leakage. The optimum arises from the simultaneous collapse of the boundary term and the rise in fidelity; longer horizons saturate in fidelity and are penalized by runtime.

still, with fidelity 0.99939 and cost 0.126 (Sec. VE); what then remains is the conditional phase of the higher levels (Sec. VII E).

The horizon dependence (Fig. 3) is more informative than the optimum itself. Very short pulses fail badly: fidelity drops to 0.726 at 40 ns and 0.882 at 50 ns, while the boundary penalties stay very large (0.399 and 0.357). Notably, the leakage remains in the same 10^{-4} band throughout this regime. The failure mode of ultra-short solutions is therefore *not* a leakage blow-up but the impossibility of satisfying accuracy and boundary regularity simultaneously under strong time compression — the speed-smoothness tension implicit in Eq. (6). This is visible sharply between 60 and 70 ns: fidelity rises from 0.982 to 0.997, the boundary term collapses from 9.75×10^{-2} to 4.28×10^{-3} , total cost falls from 0.381 to 0.158, while leakage improves only moderately.

The long-horizon side makes the multi-objective structure equally concrete. At 150 ns the fidelity is 0.99708 — slightly *better* than at the optimum — and the leakage is lower still, 6.82×10^{-5} ; yet the total cost rises to 0.285. Judged on fidelity alone the longer pulse would look preferable. Judged on the full objective it is not. The best solution is neither the most accurate nor the least leaky, but the best-balanced one.

Compared against the generic compiled reference of Eq. (12), the 70 ns Adam solution is faster by $215/70 \approx 3.1$, and against the two-CNOT synthesis of Eq. (14) by $150/70 \approx 2.1$, before any policy learning is introduced. Analog optimization on this hardware model exploits structure that a compiled implementation over a CNOT gate set cannot.

TABLE I. Complete configuration, as actually run. The four right-hand columns are the two single-target TRPO agents (Secs. IV–V), the architecture and UFO-weight sweeps (Appendix D) and the curriculum runtime sweep (Sec. VI); an entry spanning several columns applies to all of them. Every TRPO run except the nominal agent was trained with noise.

Quantity	Symbol	Single target	Arch. sweep	Weight sweep	Curriculum
<i>Physical model</i>					
Levels per mode	—			3	
Anharmonicity	η			−200 MHz	
Control bound	$u_{\max}$			20 MHz	
Filter bandwidth	B			10 MHz	
Timestep	Δt			2 ns	
Max gate duration	$T_{\max}$			180 ns	
Runtime norm.	T_{norm}			60 ns	
Target	—		$N(2.2, 2.2, \pi/2)$		$\alpha = 0, 0.157, \dots, 3.14$
<i>UFO weights</i>					
Fidelity, leakage	χ, β	10, 10	10, 10	{5, 10, 20}, 10	10, 10
Boundary	μ	0.2	0.2	{0.1, 0.2, 0.4}	0.2
Runtime	κ	0.1	0.1	{0.05, 0.1, 0.2, 0.4}	0.1
<i>TRPO</i>					
Hidden sizes	—	(64, 32, 32)	five layouts	(64, 32, 32)	(64, 32, 32)
Trust region	δ_{KL}			0.01	
Discount, GAE parameter	$\gamma_{\text{RL}}, \lambda$			0.99, 0.97	
CG iterations, damping	—			10, 0.1	
Value lr, epochs	—			10^{-3} , 50	
Initial exploration	$\log \sigma_0$	−0.5	−1.0	−1.0	−0.5
Episodes per batch	—	10 000 ^a	5000	10 000	4000
Policy iterations	—	$100 + 15 / 100 + 5^a$	20	5	≤ 150 per target
Stopping cost	C_{term}	0.40, then 0.30 ^a	0.40	0.15	0.40 ^b
Training noise	σ_{train}	0 / 1 MHz ^c	1 MHz	1 MHz	1 MHz
Seeds	—	1	1, 2, 3	1, 2	2
<i>Adam baseline (single target and family sweep)</i>					
Iterations per horizon, lr	—		400, 0.03; one deterministic initialization; seed 2		
Horizon grid	—		40–180 ns, step 10 ns		
Training noise	σ_{train}		1 MHz, one noisy trajectory per step		
Family targets	—		$\alpha = 0, 0.1, \dots, 3.1$ and π		
<i>Robustness evaluation</i>					
Dense grid	σ_{noise}	0.1–3.5 MHz, step 0.001 MHz, 60 samples; EWMA spans 50 / 80			
Matched trajectories	σ_{noise}	11 values in 0.1–3.5 MHz, 200 samples each			
Closed-loop test	σ_{noise}	1 MHz, 100 samples			

^a Both agents: iterations 1–100 with 10 000 episodes per batch and $C_{\text{term}} = 0.40$, from the same seed. They then continued with $C_{\text{term}} = 0.30$: the nominal agent for 15 iterations with 5000 episodes per batch, the noise-optimized one for 5 iterations with 10 000. The three further pairs of Sec. VC (seeds 1, 2, 3) have the settings of iterations 1–100, with the leakage bound of the noise-trained agents accumulated from the noise-free controls.

^b Also the advancement threshold. The cold-start target $\alpha = 0$ used a hand-tuned sequence of stopping costs.

^c Nominal / noise-optimized agent.

B. Nominal TRPO

Trained in the deterministic environment with the same model and weights, the best TRPO policy reaches nominal cost 0.2982, fidelity 0.99415, leakage 1.60×10^{-3} and an effective runtime of 116 ns (iteration 111). Two features deserve comment. The agent stops well before the 180 ns ceiling, confirming that it genuinely co-optimizes shape and duration rather than filling the available window. And 116 ns is below both synthesis references, by $215/116 \approx 1.85$ and $150/116 \approx 1.29$.

C. Noise-optimized TRPO and nominal comparison

The noise-optimized agent, trained under $\sigma_{\text{train}} = 1$ MHz with otherwise identical settings (Table I), reaches its lowest nominal cost at iteration 11: cost 0.3537, fidelity 0.99078, leakage 4.73×10^{-4} and runtime 98 ns. The plan that the training script saved as its best comes from iteration 91 (cost 0.3822, fidelity 0.98782, 104 ns), because a resumed segment of the run restarted the script’s best-tracking. We report the lowest-cost plan over the whole run, the rule applied to the nominal agent, and evaluate both in Sec. V. Table II collects the solutions.

The comparison that matters for the rest of the paper is the third row against the first. On this deterministic benchmark the direct optimizer beats the policy-

TABLE II. Nominal performance on $N(2.2, 2.2, \pi/2)$; best value in each column in bold. Two Adam rows are given, both from the same noise-trained horizon search: the plan selected by deterministic re-evaluation (70 ns) and the plan selected by the optimizer’s own noisy objective (60 ns). The synthesis rows are the generic reference of Ref. [4], Eq. (12), and the two-CNOT reference of Eq. (14), shown for runtime only.

Controller	T (ns)	C^{nom}	F^{nom}	L_{tot}
Adam, deterministic	70	0.1576	0.99662	2.82×10^{-4}
Adam, noisy obj.	60	0.3806	0.98205	3.67×10^{-4}
TRPO, nominal	116	0.2982	0.99415	1.60×10^{-3}
TRPO, noise-opt.	98	0.3537	0.99078	4.73×10^{-4}
Generic synthesis	215	—	—	—
Two-CNOT synthesis	150	—	—	—

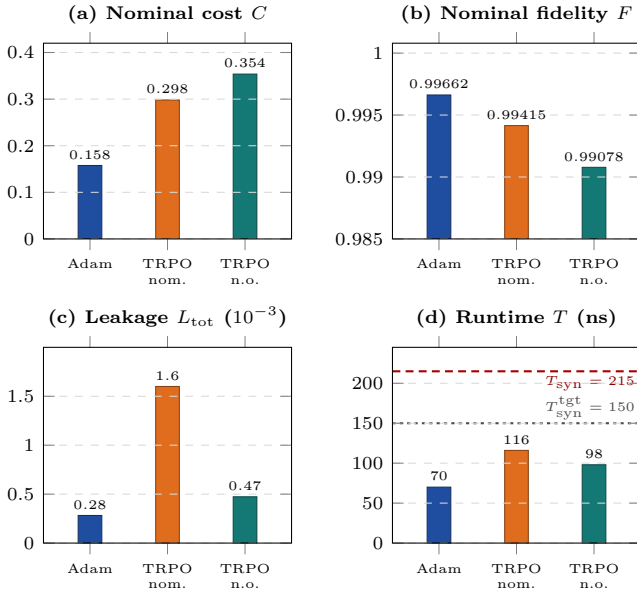


FIG. 4. Nominal comparison on $N(2.2, 2.2, \pi/2)$ of the Adam baseline at its best horizon, the nominal TRPO controller and the noise-optimized TRPO controller; values are those of Table II. (a) Total cost. (b) Nominal fidelity. (c) Total leakage. (d) Effective runtime, with the generic (215 ns) and two-CNOT (150 ns) synthesis references marked. Both TRPO controllers produce high-quality solutions, but the direct optimizer dominates both of them on all four deterministic metrics.

gradient controller simultaneously in cost ($1.89\times$), runtime ($1.66\times$), leakage ($5.67\times$) and, marginally, fidelity (2.5×10^{-3} absolute). It beats the noise-optimized agent on the same four metrics, by $2.24\times$ in cost, $1.4\times$ in runtime and $1.67\times$ in leakage, at fidelity 0.99662 against 0.99078. This is not a defect of the RL formulation; it is a delimitation of what it should be expected to deliver. When the model is known exactly and the objective is smooth and differentiable, direct gradient methods are extremely effective on fixed-target closed-system pulse design [3, 18], and nothing about policy optimization

changes that. The baseline that the robustness claims are tested against is therefore a capable one, even though its budget is small.

The noise-optimized agent’s plan sits between the two: shorter and less leaky than the nominal agent’s, but less accurate (0.99078 against 0.99415) and more expensive (0.354 against 0.298). Stochastic training has not bought nominal quality. Whether it has bought robustness is the subject of the next section.

V. ROBUSTNESS UNDER STOCHASTIC CONTROL NOISE

A. Evaluation protocol

Each stored control plan is replayed open loop under the noise model of Sec. III B on a dense grid of 3401 noise strengths spanning $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz in steps of 0.001 MHz, with 60 independent realizations at each point — about 2×10^5 noisy rollouts per controller. Sixty samples give a noisy point estimate, so the curves are read through an exponentially weighted moving average along the σ_{noise} axis (span 50 for the mean, 80 for the variance). Because neighbouring σ_{noise} values differ by 0.001 MHz, the average pools thousands of statistically comparable realizations; the underlying scatter is shown in the figures and kept in the released data, and no other post-processing is applied. Every controller is scored against $N(2.2, 2.2, \pi/2)$, with the same seed at each grid point.

Four controllers are compared: the two Adam plans of Table II (70 ns, selected by deterministic re-evaluation, and 60 ns, the optimizer’s own choice), the nominal TRPO plan (116 ns) and the noise-optimized TRPO plan (98 ns). The other noise-optimized candidate, the 104 ns plan of iteration 91, was evaluated too and lies between the two TRPO curves. Adam and the noise-optimized agent were both trained at $\sigma_{\text{train}} = 1$ MHz; the nominal agent never saw noise.

Four individual plans cannot by themselves isolate the effect of stochastic training, since each is one draw from a training process. Section V C therefore adds a controlled comparison over the whole training histories of the two TRPO agents, and Sec. V D explains what it finds.

B. Selected controllers

Figure 5 and Table III give the result. The direct optimizer is the most robust controller at every noise strength. The 70 ns Adam plan has the highest average fidelity and the lowest variance of all four up to about 2 MHz, and the shorter 60 ns plan takes over beyond (the two cross between 2.1 and 2.4 MHz in mean fidelity and between 1.9 and 2.4 MHz in variance), so that one of the two Adam plans is best on the entire grid and neither TRPO plan is best anywhere. At the training noise the 70 ns plan reaches $\bar{F} = 0.9782$, against 0.9659 for the

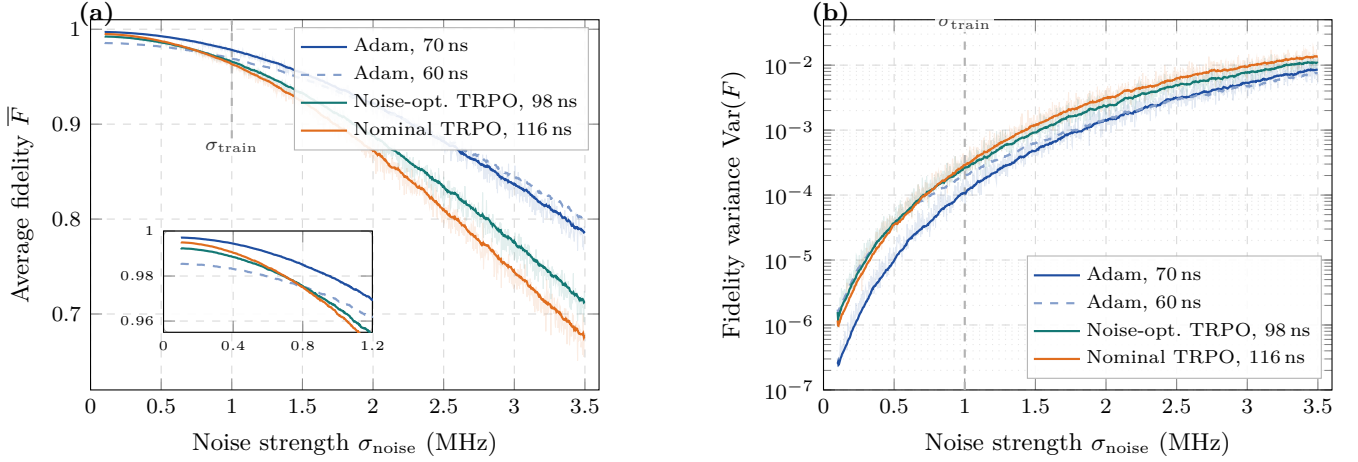


FIG. 5. Robustness on $N(2.2, 2.2, \pi/2)$ of the controllers of Table II, replayed open loop. (a) Average fidelity $\bar{F}(\sigma_{\text{noise}})$ over $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz, with the low-noise regime magnified in the inset. (b) Fidelity variance on a logarithmic axis. Solid and dashed lines are exponentially weighted moving averages along the noise axis (spans 50 and 80); the underlying Monte Carlo scatter is shown for three of the curves. The dashed vertical lines mark the training noise $\sigma_{\text{train}} = 1$ MHz. One of the two Adam plans is the most robust controller at every noise strength; at high noise the four plans rank by duration.

noise-optimized TRPO plan and 0.9631 for the nominal one.

Between the two TRPO plans the order depends on the noise strength. Below $\sigma_{\text{noise}} \approx 0.72$ MHz the nominal plan is better, as its higher nominal fidelity dictates; above, the noise-optimized plan is, with an infidelity ratio of 1.08 at 1 MHz that saturates near 1.14 at large noise. That is an order of magnitude below the gain reported in Ref. [4], and Sec. VC shows that even this much does not come from stochastic training.

A pattern in Table III anticipates the explanation. At 3.5 MHz the five evaluated plans rank exactly by duration — 60, 70, 98, 104 and 116 ns, with $\bar{F} = 0.799, 0.785, 0.713, 0.697$ and 0.676 — whichever method produced them. Over the whole sweep the fidelity lost between 0.1 and 3.5 MHz is 0.187 and 0.212 for the two Adam plans, 0.280 for the noise-optimized TRPO plan and 0.319 for the nominal one, again in order of duration.

C. Matched training histories

The two TRPO agents were trained from the same seed, with the same architecture, weights, batch size and stopping cost, for 100 iterations (Table I); the only difference was the training noise. At every iteration each agent stored the deterministic plan of its current policy. Comparing those plans iteration by iteration isolates stochastic training much better than comparing two selected plans, and the spread along the history stands in, partly, for the spread across seeds.

We evaluated the 160 plans of iterations 21–100 at 11 noise strengths between 0.1 and 3.5 MHz, with 200 realizations each. The result, Fig. 6(a), is the opposite of the intended effect. At every noise strength the noise-

TABLE III. Robustness at representative noise strengths (σ_{noise} in MHz) of the two Adam plans, the nominal TRPO plan (nom.) and the noise-optimized TRPO plan (n.o.), with their durations. Upper block: channel average fidelity $\bar{F}$, Eq. (17). Lower block: fidelity variance. All values are EWMA estimates over the dense grid; the best controller at each noise strength is in bold.

σ_{noise}	Adam 70 ns	Adam 60 ns	TRPO nom. 116 ns	TRPO n.o. 98 ns
0.1	0.9970	0.9857	0.9950	0.9925
0.5	0.9926	0.9816	0.9874	0.9861
1.0	0.9782	0.9691	0.9631	0.9659
1.5	0.9543	0.9480	0.9238	0.9325
2.0	0.9221	0.9193	0.8722	0.8883
3.0	0.8367	0.8453	0.7449	0.7761
3.5	0.7848	0.7989	0.6764	0.7128
<i>Fidelity variance $\text{Var}(F)$</i>				
0.1	2.5×10^{-7}	1.7×10^{-6}	1.2×10^{-6}	1.6×10^{-6}
1.0	1.1×10^{-4}	2.0×10^{-4}	2.9×10^{-4}	2.7×10^{-4}
2.0	1.4×10^{-3}	1.5×10^{-3}	3.1×10^{-3}	2.3×10^{-3}
3.5	8.5×10^{-3}	7.6×10^{-3}	1.4×10^{-2}	1.1×10^{-2}

optimized plan is *less* robust than the nominal plan of the same iteration in most iterations: it has the higher average fidelity in only 19–25% of them. The mean paired difference is -1.7×10^{-3} at 0.1 MHz, -3.5×10^{-3} at 1 MHz and -1.5×10^{-2} at 3.5 MHz, and its 95% block-bootstrap interval excludes zero up to 2.5 MHz. The plans of the two histories have similar durations (median 98 and 103 ns) and similar nominal fidelity (median 0.9855 and 0.9838).

Two caveats bound this result. The difference we measure is of the size of the seed-to-seed spread found in Appendix D1, so a different pair of seeds could move it by a few 10^{-3} in either direction; and the favourable com-

TABLE IV. Matched comparison for three seeds, with the leakage bound of the noise-trained agents taken from the noise-free controls; iterations 21–100. Fraction of iterations in which the noise-trained plan has the higher $\bar{F}$ at 1 MHz; mean paired difference in $\bar{F}$ (noise-trained minus nominal) at three noise strengths; mean difference in duration and in nominal fidelity; and the difference at 1 MHz that Eq. (21) predicts from those two. The last column is the pair of Fig. 6(a).

	Seed 1	Seed 2	Seed 3	First pair
Noise-trained better, 1 MHz	26%	28%	24%	21%
$\Delta\bar{F}$ at 0.1 MHz (10^{-3})	-0.5	+0.1	-0.7	-1.7
$\Delta\bar{F}$ at 1 MHz (10^{-3})	-1.6	-1.7	-2.0	-3.5
$\Delta\bar{F}$ at 3.5 MHz (10^{-3})	-11.5	-11.9	-12.1	-14.9
ΔT (ns)	+5.1	+5.7	+5.2	—
ΔF^{nom} (10^{-3})	-0.5	+0.1	-0.9	—
Predicted $\Delta\bar{F}$ at 1 MHz (10^{-3})	-1.9	-1.6	-2.2	—

parison of the two selected plans above shows how easily one pair of plans can suggest the opposite. What the data do rule out is an improvement of the size reported in Ref. [4]: an order of magnitude in infidelity at 1 MHz would mean $\bar{F} \approx 0.996$ instead of 0.96, far outside the spread of either history.

Three seeds, with the leakage bound taken from the noise-free controls. The noise-optimized agent above was trained with a leakage bound that responds to the noise itself (Sec. V D), and it is one seed. We therefore trained three further pairs of agents, with seeds 1, 2 and 3 and the settings of the matched iterations (100 iterations of 10 000 episodes, $C_{\text{term}} = 0.40$). In these the noise-trained agent accumulates the leakage bound from the noise-free controls, so that the training noise enters its objective through the fidelity alone. The runs use a batched implementation of the environment, which agrees with the original one to 10^{-13} in the cost at every step but draws different random numbers, so its seed 1 is not the pair above. The 600 plans were evaluated exactly as before (Table IV).

The result is the same. From 0.75 MHz up, the noise-trained plan is the more robust one in 17–34% of the iterations, in every seed and at every noise strength, and the 95% interval of the paired difference excludes zero in all three seeds; below, it does in two seeds, while in the third the two agents are level. The difference is about half that of the first pair: taking the noise out of the leakage bound reduces the handicap of the noise-trained agent and does not turn it into an advantage. Its size remains that of the spread between seeds. Pairing each noise-trained history with each nominal one, whatever their seeds, gives differences between -3.5 and $+0.2 \times 10^{-3}$ at 1 MHz, so what three seeds establish is the absence of a gain, not a loss intrinsic to stochastic training. Nor is the difference a matter of pulse shape. The noise-trained plans are 5–6 ns longer on average and differ in nominal fidelity by 10^{-3} or less, and Eq. (21) turns those two numbers into -1.6 to -2.2×10^{-3} at 1 MHz, against -1.6 to -2.0×10^{-3} measured. No plan of the six histories

exceeds $\bar{F} = 0.968$ at 1 MHz.

D. Why stochastic training does not reach the deployed pulse

Three observations explain the null result.

First, under this noise model the pulse shape cannot buy robustness. This is the white-noise limit of the filter-function description of open-loop control [20–22]: the response of a pulse to noise is the overlap of the noise spectrum with a filter function fixed by the pulse, and a flat spectrum sees only the integral of that function, which the pulse shape does not change (Appendix C). In the present model the argument is short. The noise is independent from one 2 ns step to the next. Moving a kick to the end of the pulse conjugates its generator G by the evolution so far, and as long as that evolution keeps the computational subspace invariant, the first-order infidelity of the kick depends only on the variance $V(G) = \text{Tr}(PG^2P)/d - |\text{Tr}(PGP)/d|^2$ of G over that subspace P , which conjugation leaves unchanged. To leading order the noise-induced infidelity is therefore independent of the pulse shape and proportional to the duration,

$$1 - \bar{F}(\sigma_{\text{noise}}) \simeq 1 - \bar{F}(0) + K\sigma_{\text{noise}}^2 T, \\ K = \frac{d}{d+1} (2\pi \cdot 10^{-3})^2 \Delta t \sum_c V_c, \quad (21)$$

with σ_{noise} in MHz, Δt and T in ns, and the sum running over the noisy channels. The components of G that lead out of the subspace, to states detuned by the anharmonicity, are partly averaged during the step; we weight them by the zero-order-hold factor $\text{sinc}^2(\omega\Delta t/2)$ at their transition frequency ω . For the gmon model this gives $V = 1/4$ for each detuning, 1.57 for each drive, 1.07 for the coupling and 0 for the anharmonicity, and hence $K = 2.98 \times 10^{-4} \text{ ns}^{-1} \text{ MHz}^{-2}$ (3.79×10^{-4} without the step averaging). The measured excess infidelity per unit duration at 1 MHz is 2.97×10^{-4} and 2.99×10^{-4} for the plans of the nominal and noise-optimized histories, with interquartile ranges within $\pm 2\%$; at 0.5 and 2 MHz it is 3.4 and 2.7×10^{-4} , the drift reflecting higher orders and the approximate treatment of the leaking components. Evaluated without the sinc approximation, with the exact response of every step (Appendix C), the first-order rate of the 160 plans of the two histories lies between 2.9794 and 2.9805×10^{-4} : the pulse shape moves it by four parts in 10^4 . Two thirds of it come from the two drive amplitudes, 23% from the coupling and 11% from the detunings, and replaying the pulses with the noise on one channel at a time reproduces each share (Table VII). The value of K matters less than the fact that it is the same for every plan: excess infidelity and duration are correlated at 0.99 across both histories, and the two Adam plans fall on the same line (Fig. 6(c)). Against white control noise an open-loop pulse is robust

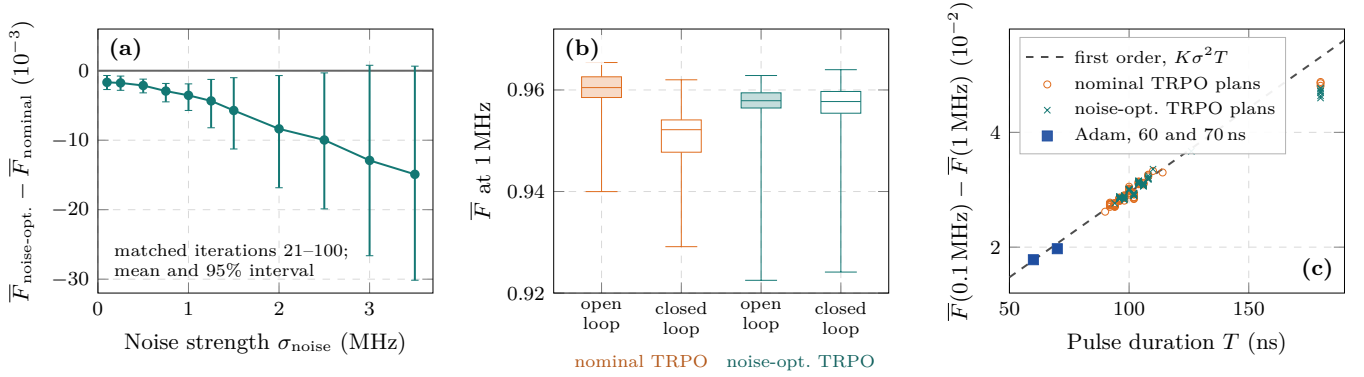


FIG. 6. What stochastic training changes. (a) Paired difference in average fidelity between the noise-optimized and the nominal TRPO plan of the same iteration, over the matched iterations 21–100, with 95% block-bootstrap intervals; negative values favour the nominal agent. (b) Average fidelity at 1 MHz of the same checkpoints, replayed open loop (filled) and acting in closed loop on the noisy propagator (open); boxes span the quartiles and whiskers the full range. (c) Excess infidelity at 1 MHz against pulse duration for all plans of both histories and for the two Adam plans, with the first-order prediction of Eq. (21) (dashed); the plans at 180 ns are rollouts that reached the episode ceiling, where higher orders begin to matter.

only by being accurate and short. Every difference in Fig. 5 follows from those two numbers, and a noise-aware objective can improve robustness only through them.

Second, the training signal is dominated by the noise itself. Every term of Eq. (8) is evaluated on the noisy trajectory, including the leakage bound, whose second time derivative responds to the step-to-step jumps of white noise. At $\sigma_{\text{train}} = 1$ MHz the bound grows by 3.8×10^{-3} per ns for every plan we replayed, about 27 times the leakage population actually present at the end of the pulse; for the 116 ns nominal plan the term βL_{tot} is 4.5, against 0.50 for $\chi(1 - F)$. In the noisy objective, leakage thus acts as an implicit runtime penalty about 23 times stronger than the explicit one, κ/T_{norm} , and one that is nearly blind to pulse shape. This explains two observations. The Adam horizon search, whose sampled losses rise by about 0.04 per ns from 60 ns on, prefers the 60 ns plan to the more accurate 70 ns one. And the cost of a noisy TRPO rollout never falls below the stopping threshold: all 1.05×10^6 training episodes of the noise-optimized agent ran to the 180 ns ceiling, so the co-optimization of stopping time that distinguishes the RL formulation was inactive throughout noisy training. The bound is not the only reason for the latter. In the three noise-trained agents of Table IV, whose bound does not see the noise, the training episodes still lasted 180.0 ns on average, and those of the nominal agents 177–178 ns: the exploration noise of the stochastic policy, with the training noise on top of it, keeps the cost of a training rollout above a threshold that the deterministic plan of the same policy reaches at about 100 ns.

Third, what stochastic training changes is the policy’s response to noisy states, and that response is not deployed. During training the agent observes the noisy propagator and acts on it. We ran every checkpoint of both histories in the noisy environment at 1 MHz, in closed loop and for the duration of its own plan,

and compared it with the open-loop replay of that plan (Fig. 6(b)). In closed loop the noise-optimized agent *is* the more robust one: it beats the nominal agent in 81% of iterations, by 5.4×10^{-3} in $\bar{F}$ on average (95% interval $[3.4, 7.4] \times 10^{-3}$). The reason is that the nominal agent, fed states it never saw in training, reacts badly: its closed-loop fidelity falls below its own open-loop value in 99% of iterations, by 8.8×10^{-3} on average, with three times the variance. Stochastic training removes this instability — the noise-optimized agent’s closed-loop and open-loop fidelities agree within 10^{-3} — but it neither produces feedback that beats open loop nor changes the open-loop pulse. A robustness test carried out in the training environment would credit the noise-optimized agent with an advantage that its deployable pulses do not have.

E. Noise with memory, and the same test with the direct optimizer

Equation (21) makes two predictions that can be tested without retraining an agent. The first is that the pulse shape should matter as soon as the noise is correlated in time; the second, that a noise-aware optimizer should then be able to use it, and should gain nothing under white noise.

Existing pulses. We replayed every stored plan under quasi-static noise, one Gaussian offset per channel held for the whole pulse and redrawn for every shot, and evaluated Eq. (C2) for it (Appendix C; the two agree within 3% at 0.1 MHz). At equal σ_{noise} the infidelity is 15 to 23 times the white-noise one for the controllers of Table II, because the offsets of successive steps add coherently, and it grows as T^2 instead of T [Fig. 7(c)]. Short pulses gain even more than under white noise: at 0.1 MHz the excess infidelity is 2.6×10^{-3} and 3.6×10^{-3} for the 60

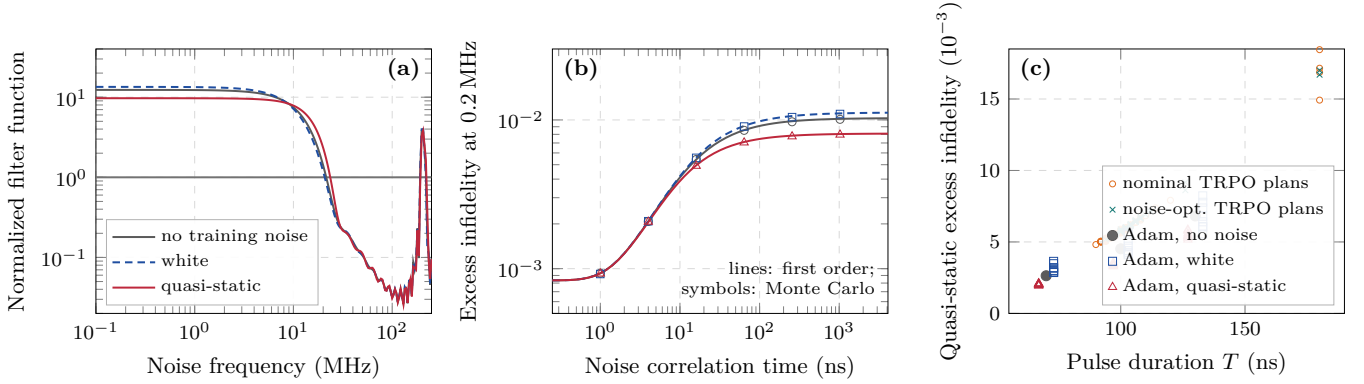


FIG. 7. Noise with memory. (a) Filter functions, summed over the channels and normalized to their average over the Nyquist band, of three 70 ns Adam pulses for $N(2.2, 2.2, \pi/2)$: optimized without noise, under white noise (1 MHz) and under quasi-static noise (0.3 MHz). The average is the same for the three (horizontal line); the low-frequency plateau is not. The peak at 200 MHz is the transition to the leakage levels. (b) Excess infidelity of the same pulses under exponentially correlated noise of standard deviation 0.2 MHz against its correlation time: first-order prediction of Eq. (C2) (lines) and Monte Carlo replay (8000 realizations, symbols). (c) Excess infidelity under quasi-static noise of 0.1 MHz against duration, for the plans of both TRPO histories and for Adam pulses optimized at 70, 100 and 130 ns under the three noise models (eight seeds for each stochastic model; the white and quasi-static symbols are shifted by ± 3 ns for visibility).

and 70 ns Adam plans, against 5.6×10^{-3} and 8.0×10^{-3} for the noise-optimized and nominal TRPO plans. The shape now enters, but among pulses that were not optimized for it, only weakly: divided by T^2 , the sensitivity of the 160 plans of the two TRPO histories spans 5.7 to $6.3 \times 10^{-7} \text{ ns}^{-2}$, and the noise-optimized agent is not ahead. It has the higher average fidelity in 25% of the matched iterations at 0.1 MHz and 20% at 0.3 MHz, and its sensitivity per T^2 differs from the nominal agent's by $0.3 \times 10^{-8} \text{ ns}^{-2}$ (95% interval $[-0.6, 1.2] \times 10^{-8}$). Training under white noise has not prepared the policy for noise with memory either.

The direct optimizer under three noise models. We then repeated the comparison of Sec. V C with the gradient optimizer, where it costs minutes. At a fixed horizon, so that the duration cannot change, the pulse was optimized (i) on the deterministic objective, (ii) under the white training noise of the paper, $\sigma_{\text{train}} = 1$ MHz, and (iii) under quasi-static noise of 0.3 MHz, one offset per channel and per trajectory, a single-sample version of ensemble-based robust control [23]; eight seeds for each stochastic model, horizons of 70, 100 and 130 ns, otherwise the settings of Sec. III C. Table V gives the result.

Under white noise the three models give the same first-order sensitivity to four digits, as Eq. (21) requires, and the only thing that white-noise training changes is the nominal fidelity, for the worse: none of the 24 white-trained pulses is more accurate, or more robust against the very noise it was trained on, than the pulse optimized without noise at the same horizon. Its stochastic gradient costs $5\text{--}10 \times 10^{-3}$ in nominal fidelity and buys nothing. The null result of Sec. V C is therefore not a peculiarity of the two TRPO runs: it recurs, with seeds, in an optimizer whose gradients are exact.

Part of that cost is specific to our objective, in which

TABLE V. Adam at fixed horizon T under three training-noise models (mean over eight seeds for the stochastic ones; the noise-free optimization is deterministic). $1 - F$: nominal infidelity. White: first-order excess infidelity at 1 MHz, Eq. (C2), and average fidelity $\bar{F}$ from 2000 realizations. Quasi-static: first-order excess infidelity at 0.1 MHz, and $\bar{F}$ at the training strength of 0.3 MHz. Rows marked *: the leakage bound of the training objective is accumulated from the noise-free controls instead of the noisy Hamiltonian; under that choice the quasi-static rows change by less than 10^{-5} in every fidelity.

T (ns)	Training noise	$1 - F$ (10^{-3})	White, 1 MHz		Quasi-static	
			excess (10^{-2})	$\bar{F}$	excess (10^{-3})	0.3 MHz $\bar{F}$
70	none	0.6	2.085	0.9792	2.57	0.9762
	white	5.4	2.085	0.9757	3.04	0.9682
	white*	2.6	2.085	0.9776	2.79	0.9727
	quasi-static	3.5	2.086	0.9767	2.01	0.9792
100	none	0.3	2.980	0.9707	4.43	0.9602
	white	5.4	2.980	0.9667	4.73	0.9541
	white*	3.7	2.980	0.9682	4.50	0.9569
	quasi-static	5.9	2.980	0.9663	3.38	0.9654
130	none	0.2	3.874	0.9624	6.71	0.9411
	white	10.3	3.874	0.9546	6.75	0.9334
	white*	8.2	3.874	0.9567	6.69	0.9346
	quasi-static	16.4	3.874	0.9504	5.47	0.9393

the leakage bound is accumulated from the noisy Hamiltonian and responds to the noise itself (Sec. V D). We therefore repeated the experiment with the bound taken from the noise-free controls, so that the noise enters the objective through the fidelity alone (rows marked * in Table V). The cost of white-noise training in nominal fidelity falls from 4.8, 5.1 and 10.1×10^{-3} to 2.0, 3.5 and 8.1×10^{-3} at the three horizons, and the outcome is the

same: again none of the 24 pulses is more accurate, or more robust against white noise, than the one optimized without noise. Over the two variants white-noise training helped in none of 48 runs.

Quasi-static training does what white training cannot. It lowers the quasi-static sensitivity in every one of the 24 runs, by 18–24% on average and with a spread of a few percent between seeds, by moving weight of the filter function away from zero frequency while leaving its average untouched [Fig. 7(a)]; the three pulses coincide for correlation times below a few nanoseconds and separate beyond [Fig. 7(b)]. The gain is modest, because a 70–130 ns pulse through a 10 MHz filter has little room for the refocusing structure of dynamically corrected gates [24], and it is paid for in nominal fidelity, so it shows in $\bar{F}$ only once the noise is strong enough: at 0.1 MHz a single quasi-static-trained pulse beats the noise-free one, at the training strength of 0.3 MHz 19 of 24 do, against 2 of 24 white-trained ones.

F. Interpretation

The robustness mechanism proposed in Ref. [4] does not survive a controlled test in our implementation. Two agents that differ only in the training noise produce open-loop pulses of the same robustness, and the direct optimizer, trained under the same noise, produces the most robust pulses of all, because they are the shortest and among the most accurate. Equation (21) shows that this is not an accident of our optimizer settings: against temporally uncorrelated amplitude noise, nominal fidelity and duration determine open-loop robustness to leading order, whatever method produced the pulse. Robustness that depends on the pulse shape requires noise with memory — quasi-static offsets, drift or low-frequency spectra — which is what robust-control techniques are designed for [3] and what the UFO noise model does not contain. Section V E shows both halves of this statement in the same optimizer: trained against quasi-static noise it lowers the sensitivity of its pulses by about a fifth, trained against white noise it lowers nothing.

VI. RUNTIME ACROSS THE GATE FAMILY

A. Protocol

To test hardware efficiency beyond a single target we sweep the subfamily

$$U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2), \quad \alpha \in [0, \pi], \quad (22)$$

whose structure was stated in Sec. II F, in 21 steps of 0.157. Rather than retraining from scratch at each point, the best checkpoint obtained at one α initializes training at the next — a curriculum that exploits continuity in target space, in the spirit of the transfer strategy of

Ref. [4]. Training at each target stops when the cost falls below 0.40 or after 150 iterations, and we call a target *converged* when its final deterministic rollout reaches that cost. The environment is stochastic at $\sigma_{\text{train}} = 1$ MHz. The first target, $\alpha = 0$, is a cold start trained from scratch with a hand-tuned sequence of stopping costs.

As a no-transfer control we also sweep the same family with the Adam baseline, optimizing each of 33 targets ($\alpha = 0, 0.1, \dots, 3.1$ and π) independently over the horizon grid of Table I, with the horizon chosen by the noisy training objective as in Sec. IV A. This separates two effects that the curriculum conflates: how the difficulty of the control problem varies across the family, and what the curriculum’s history adds to it.

B. Runtime landscape of the curriculum

Table VI and Fig. 8 summarize the curriculum sweep. Of the 21 targets, 14 converge. One of them is degenerate: at $\alpha = 1.570$ the target is the identity to within $|\varepsilon| = 8 \times 10^{-4}$, Eq. (13), so the cost is already below threshold after the first 2 ns step and the phase ends after one iteration with fidelity 0.999999 — the value that doing nothing at all would give. It is not a fast gate, and we exclude it from every aggregate below.

The 13 non-trivial converged targets take 76–148 ns (median 132 ns), at nominal fidelity 0.982–0.992 and $\bar{F} = 0.950$ –0.967 at $\sigma_{\text{noise}} = 1$ MHz. The other 7 do not converge: the cold start $\alpha = 0$, the interval $\alpha \in [0.47, 0.79]$ and the edge $\alpha \geq 2.83$. Their final rollouts run to the 180 ns ceiling at nominal fidelity 0.92–0.95 (0.56 at $\alpha = 0$), and the lowest cost reached anywhere along them, 0.43–0.80 at 142–174 ns, stays above the threshold. They also carry somewhat larger leakage (Fig. 8(c)). These are failures to converge within the budget, not slow gates.

The mirror symmetry of Sec. II F shows that the failures belong to the optimizer rather than to the targets. The curriculum grid is symmetric about $\pi/2$ to within 0.002, and in five of its ten mirror pairs one side converges while the other fails. Reflecting the converged pulses ($g \rightarrow -g$, $\phi_1 \rightarrow \phi_1 + \pi$) reproduces their cost to 10^{-12} on the mirrored targets. For instance, the 118 ns pulse for $\alpha = 2.355$ implements $\alpha = \pi - 2.355$ with fidelity 0.986, where the curriculum itself ended at 180 ns and 0.949. Only the two ends of the interval, both close to the local gate $\sigma_1^z \sigma_2^z$, fail on both sides. The converged runtimes do not follow the target structure either: their rank correlation with $|\varepsilon|$ is 0.51 ($p = 0.08$), and with the Weyl coordinate a it is 0.01. The curriculum landscape records the order in which targets were visited and the stopping rule (Sec. VI D) more than the targets themselves.

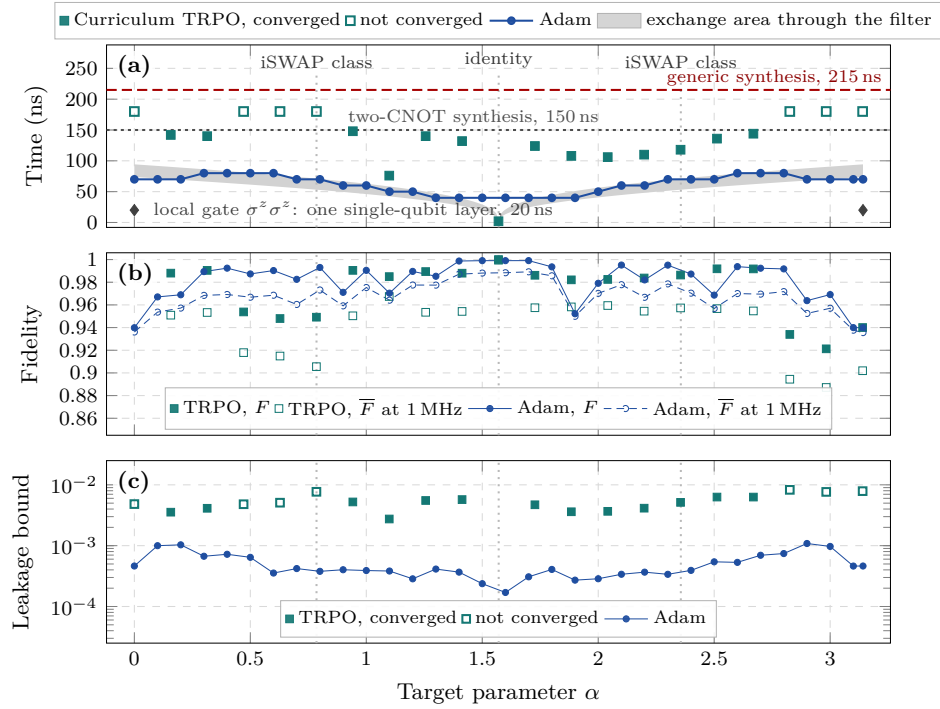


FIG. 8. Runtime across $N(\alpha, \alpha, \pi/2)$. (a) Final evaluation runtime of the curriculum controller (filled squares: converged; open squares: not converged, the rollout running to the 180 ns ceiling) and horizon selected by the Adam baseline (circles). Grey band: shortest time in which the coupling can deliver the exchange area $2|\alpha - \pi/2|$ through the bandwidth filter under the 20 MHz bound, without (lower edge) and with (upper edge) the condition that it returns to zero at the end of the pulse (Sec. VIE). Dashed: generic synthesis reference of Ref. [4], 215 ns; dotted: two-CNOT synthesis, Eq. (14), 150 ns; diamonds: the local members $\sigma_1^z \sigma_2^z$, one layer of single-qubit gates. Vertical lines mark the identity target $\alpha = \pi/2$ and the iSWAP-equivalent targets $|\alpha - \pi/2| = \pi/4$. (b) Nominal fidelity and average fidelity at $\sigma_{\text{noise}} = 1$ MHz for both sweeps; the cold-start curriculum point $\alpha = 0$ (0.56 and 0.61) is off scale. (c) Leakage bound of the evaluated plans.

C. Comparison with gate synthesis

Measured against the generic reference $T_{\text{syn}} = 215$ ns, every converged non-trivial target is faster, by $1.45\text{--}2.83\times$ (median $1.63\times$). This is the comparison of Ref. [4], and it needs two qualifications. First, the 180 ns episode cap lies below 215 ns, so even the failed rollouts would formally beat the reference; a speedup is meaningful only for a pulse that implements the gate. Second, the reference is generic. Against the target-specific time of Eq. (14), 150 ns for every non-local member, the converged curriculum pulses are faster by only $1.01\text{--}1.97\times$ (median $1.14\times$), and the local ends of the family, a single layer of single-qubit gates in a circuit, are exactly where the curriculum fails.

The order-of-magnitude reduction reported in Ref. [4] refers to $\alpha \in [1.2, 1.7]$. By Eq. (13) this interval is the neighbourhood $|\epsilon| \leq 0.37$ of the identity, where the required exchange area can be accumulated in about 6 ns at the coupling bound. A large ratio to a fixed three-CNOT reference is to be expected there from any working pulse optimizer; it reflects the reference more than the controller. In our sweep the only target in that interval with a ratio of order ten is the identity itself.

None of this removes the physical point. The synthesis benchmark is optimal within a circuit model over a fixed gate library, whereas the analog controller optimizes in the continuous Hamiltonian-control space, and the two solve related but different problems. Nothing in these results says that a quantum speed limit has been beaten. What they do say is that the size of the analog advantage depends on the reference, and that a fair reference must be target-specific — or, on hardware with a native tunable exchange gate [16], must be that gate.

D. What the runtime comparison measures

The independently optimized Adam baseline is faster than the curriculum controller across the family. Its selected horizons span 40–80 ns with a median of 70 ns (Fig. 9), against a median of 132 ns for the converged curriculum targets, at comparable nominal fidelity (median 0.987 against 0.988) and better robustness at $\sigma_{\text{noise}} = 1$ MHz (median $\bar{F} = 0.969$ against 0.955) — the latter as Eq. (21) predicts for shorter pulses. Against Eq. (14) the 31 non-local Adam solutions are faster by $1.9\text{--}3.75\times$ (median $2.1\times$), with the same range and median for the

TABLE VI. Curriculum runtime sweep over $N(\alpha, \alpha, \pi/2)$. $\varepsilon = \alpha - \pi/2$ is the exchange angle of Eq. (13) and T the final evaluation runtime; the last two columns give the speedup over the generic reference, 215 ns, and over the target-specific synthesis time of Eq. (14), 150 ns for every row shown. Rows marked † did not converge (final cost above 0.40): their roll-outs run to the 180 ns ceiling and no speedup is quoted. The row marked ‡ is the identity target.

α	ε	T (ns)	F^{nom}	$\bar{F}_{\sigma=1}$	215/ T	$T_{\text{syn}}^{\text{tgt}}/T$
0.000†	-1.571	180	0.5577	0.6133	—	—
0.157	-1.414	142	0.9880	0.9510	1.51	1.06
0.314	-1.257	140	0.9905	0.9533	1.54	1.07
0.471†	-1.100	180	0.9538	0.9179	—	—
0.628†	-0.943	180	0.9480	0.9149	—	—
0.785†	-0.786	180	0.9492	0.9055	—	—
0.942	-0.629	148	0.9905	0.9504	1.45	1.01
1.099	-0.472	76	0.9850	0.9674	2.83	1.97
1.256	-0.315	140	0.9894	0.9535	1.54	1.07
1.413	-0.158	132	0.9880	0.9542	1.63	1.14
1.570‡	-0.001	2	0.99999	0.9994	—	—
1.727	0.156	124	0.9861	0.9575	1.73	1.21
1.884	0.313	108	0.9821	0.9582	1.99	1.39
2.041	0.470	106	0.9825	0.9595	2.03	1.42
2.198	0.627	110	0.9837	0.9545	1.95	1.36
2.355	0.784	118	0.9863	0.9573	1.82	1.27
2.512	0.941	136	0.9919	0.9567	1.58	1.10
2.669	1.098	144	0.9918	0.9547	1.49	1.04
2.826†	1.255	180	0.9340	0.8944	—	—
2.983†	1.412	180	0.9213	0.8872	—	—
3.140†	1.569	180	0.9400	0.9020	—	—

21 of them whose nominal fidelity exceeds 0.98; at the two local ends Adam’s 70 ns pulses, with fidelity 0.940, are slower than a 20 ns layer of single-qubit gates. Three caveats apply to the Adam numbers: the horizon grid starts at 40 ns, so the seven targets that sit at the grid minimum could be faster still; the horizon is selected by the noisy objective, which favours short pulses (Sec. V D); and nine of the 33 solutions have nominal fidelity below 0.97. The synthesis times, moreover, are those of ideal gates with no infidelity attached, whereas the pulses set against them have fidelities of 0.98–0.99.

The runtime advantage over compiled synthesis is therefore a property of *direct analog optimization on this hardware model*, not a benefit conferred by reinforcement learning.

The curriculum runtimes are, in addition, partly an artifact of the stopping rule. Training at each target halts as soon as the UFO cost drops below the advancement threshold 0.40, and every converged phase ends in the narrow band $C \in [0.393, 0.400]$. The agent stops improving the moment it becomes good enough to advance and has no incentive to shorten the pulse further, so the curriculum runtimes are times sufficient to reach a fixed cost under a bounded budget, not the shortest times the policy could achieve. A tighter threshold would very likely move the curve down; we did not run that sweep.

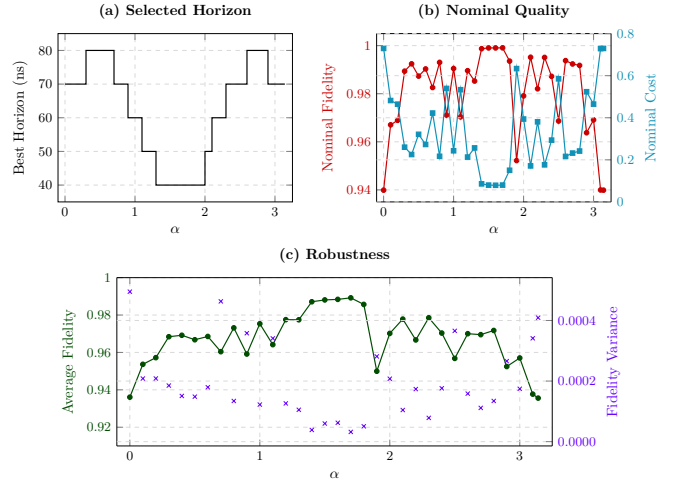


FIG. 9. No-transfer control: the Adam baseline swept independently over 33 targets of $N(\alpha, \alpha, \pi/2)$. (a) Selected horizon versus α . (b) Nominal fidelity and cost. (c) Average fidelity at $\sigma_{\text{noise}} = 1$ MHz and the corresponding variance. The horizon is shortest around the identity target $\alpha = \pi/2$ and grows with the exchange angle $|\alpha - \pi/2|$.

E. Physical interpretation

Equation (13) gives the family a single physical coordinate, the exchange angle $|\varepsilon| = |\alpha - \pi/2|$ that the coupling must accumulate. The direct optimizer, which solves each target independently, follows it closely. Its horizon is 40 ns for all seven targets with $|\varepsilon| \leq 0.33$, rises to 50–60 ns for $|\varepsilon| \in [0.37, 0.67]$ and to 70–80 ns beyond, and drops back to 70 ns only at the two ends, where the target approaches a local gate; the rank correlation is 0.80 ($p < 10^{-7}$), and 0.97 for $|\varepsilon| \leq 1.3$. It does not follow the entangling content: the iSWAP-equivalent members, the most entangling of the family, sit at 70 ns like the local ends, and the rank correlation with the Weyl coordinate a is 0.27 ($p = 0.14$).

The mechanism is the one the Hamiltonian suggests. The native coupling generates every member directly, Eq. (13), so what varies across the family is not how well the target aligns with the hardware — every member aligns perfectly — but how much exchange area the pulse must accumulate through a bandwidth-limited channel that must also start and end at zero. Near $\alpha = \pi/2$ the target is close to the identity and the pulse can be short and gentle; toward the ends it must be longer, and at the ends a local-gate alternative appears that the optimizer exploits only partly. The favourable central region of both sweeps is therefore the neighbourhood of the identity, not a basin of especially hardware-friendly entanglers.

This can be made quantitative. The filter (15) is linear, so the largest exchange area $\int g dt$ that a control bounded by 20 MHz can deliver in N steps is a linear program, with or without the condition that the filtered coupling is back at zero at the last step, which the boundary term of

Eq. (8) enforces softly. Requiring the area $2|\varepsilon|$ of Sec. III F gives the band of Fig. 8(a): from 28–40 ns at $|\varepsilon| = 0.16$ to 74–96 ns at the ends of the family. It is an estimate for pulses that use the coupling alone, not a speed limit of the model; near the ends the optimizer also uses the detunings. The horizons selected by the direct optimizer lie inside the band for 20 of the 26 targets that are not pinned at the 40 ns grid minimum, and one grid step above it for one more. For $|\varepsilon| \leq 1.03$ its pulses deliver 75–99% of the required area with the coupling; beyond, the detunings take over. The exceptions are the five targets with $|\varepsilon| \geq 1.43$, where the selected 70 ns is shorter than even the lower edge: there the pulses are detuning pulses whose area, 2.77–2.80, falls short of the π that the local gate needs, which is why their fidelity stops at 0.94–0.97. The converged curriculum pulses are 1.3–3.3 times longer than the upper edge (median 1.8). Within this model, then, the direct optimizer runs at the bandwidth limit and the learned pulses do not.

The same linear program applies to the single-qubit gates of the synthesis reference. A rotation by π needs a drive area $\pi/2$, which the filtered, bounded drive delivers in no less than 54 ns, or 72 ns if it must end at zero, and a $\pi/2$ rotation needs 40 and 56 ns, against the 20 ns per layer assumed in Eqs. (12) and (14). Both synthesis references therefore belong to hardware with a wider control bandwidth than the model in which the pulses are optimized, and the ratio between the two measures that mismatch as much as it measures analog control.

VII. DISCUSSION

A. What reproduces

The framework itself reproduces. Rebuilt from the published description — multilevel gmon model, second-order TSWT leakage bound, bandwidth filter, UFO objective, TRPO in a stochastic environment — it produces genuine analog two-qubit gates: nominal fidelity around 0.99 with leakage bounds of 10^{-4} – 10^{-3} , in pulses shorter than the compiled reference. The original UFO weights sit close to the best region of an independent weight sweep, and the original network layout is as good as any we tried within the seed-to-seed spread. The curriculum strategy works in the sense Ref. [4] describes: once warm-started, seven of the thirteen non-trivial converged targets were solved within 20 iterations.

The absolute fidelities are lower than those reported. Ref. [4] quotes gate infidelities of order 10^{-3} for every member of the family (its Fig. 3); our 13 non-trivial converged targets have nominal infidelities of 0.8 – 1.8×10^{-2} , and 7 of the 21 targets do not converge (Sec. VI). On $N(2.2, 2.2, \pi/2)$ the gap is smaller: at the lowest noise strength the average fidelity is 0.995 there, against 0.9950 and 0.9925 for our nominal and noise-optimized agents (Table III). We have not identified the cause; the training budget, the single seed and the coarser time step are

candidates.

The qualitative runtime claim also holds: analog pulses on this hardware model are shorter than a CNOT-based synthesis of the same targets, and the advantage is largest near the centre of the family.

B. What does not

Three claims are not supported in our implementation, and a fourth needs a different reading.

Nominal superiority. On $N(2.2, 2.2, \pi/2)$ a single-start Adam optimization beats both TRPO agents on cost, runtime, leakage and fidelity (Table II). The SGD baseline of Ref. [4] was also Adam with a noisy objective, and it received about 2000 random restarts, yet its average fidelity on this same target stays near 0.75–0.77 over the whole noise range of its Fig. 4. We cannot reconstruct why; the gap between the two gradient baselines is larger than any gap between RL and gradient optimization that we observe. With a gradient optimizer that works, the reported two orders of magnitude in infidelity over SGD do not reproduce: at every noise strength our direct optimizer is ahead of the RL controllers. This is the least general of our findings. It sets one gradient optimizer against our TRPO agents, which reach cost 0.30–0.35 where the optimizer reaches 0.16 and whose curriculum fails on a third of the family (Sec. VI); a better-trained agent could narrow it.

Robustness from stochastic training. In a controlled comparison, noise-aware training does not make the deployed pulse more robust (Sec. VC), and Eq. (21) explains why it could not: against per-step white noise the open-loop robustness of a pulse is fixed, to leading order, by its nominal fidelity and its duration. An order-of-magnitude difference in infidelity between two open-loop pulses of similar fidelity and length cannot arise under this noise model. Differences of that size can only come from different nominal fidelities or durations, or from an evaluation in which the policy acts on the noisy state (Sec. VD); Ref. [4] does not give enough detail to tell which, if either, applies to its results.

The runtime advantage as a property of RL. The independently optimized direct baseline is faster than the curriculum controller across the family (Sec. VID); the analog speedup belongs to pulse-level optimization, not to the learning method.

The order-of-magnitude runtime reduction. It is located at $\alpha \in [1.2, 1.7]$, around the target that is the identity, and it is measured against a generic three-CNOT circuit. Against a target-specific two-CNOT synthesis the converged curriculum pulses are 1.0–2.0 $\times$ faster and the direct optimizer 1.9–3.75 $\times$; the local members of the family need no entangling gate at all.

C. Comparing noise scales

One further point limits any quantitative comparison with Ref. [4]. Because the kicks are independent from step to step, the noise power per unit bandwidth is $\sigma_{\text{noise}}^2 \Delta t$, and K in Eq. (21) is proportional to Δt : the same σ_{noise} describes a different physical noise level at a different time step, and it matters whether amplitudes are quoted in cyclic or angular units. Replaying our pulses with the noise redrawn every 0.1, 0.5, 1, 4 or 8 ns instead of every 2 ns, at the same σ_{noise} , multiplies the excess infidelity by 0.06, 0.31, 0.6, 1.3 and 2.5, as the first-order rate predicts to within 5%. It also matters where the noise enters. Added to the raw controls, before the bandwidth filter instead of after it, the same σ_{noise} costs our five pulses 5 to 18 times less infidelity, and reaches the Hamiltonian with the 32 ns memory of the filter. In physical terms, the detuning noise of our model at $\sigma_{\text{noise}} = 1$ MHz and $\Delta t = 2$ ns is white frequency noise that dephases each qubit in $T_\varphi = 2/[(2\pi\sigma_{\text{noise}})^2\Delta t] \approx 25 \mu\text{s}$, and 3.5 MHz corresponds to $2 \mu\text{s}$.

Ref. [4] describes its pulses as piecewise constant with “around one thousand time steps” each, which for gates of about 100 ns is a step of about 0.1 ns, twenty times shorter than ours; it does not state its convention for σ_{noise} . At that step the same σ_{noise} costs sixteen times less infidelity, and the fidelity loss that its Fig. 4 shows over $\sigma_{\text{noise}} \in [0.1, 3.5]$ MHz is indeed more than an order of magnitude smaller than the loss we obtain over the same nominal range for pulses of comparable length. There the average fidelity of the noise-optimized controller stays between 0.995 and 0.98, within 0.007 of the ideal at 1 MHz; ours falls from 0.9925 to 0.713, with 0.966 at 1 MHz (Table III). Dividing our loss relative to 0.1 MHz by sixteen gives about 0.991 at 1 MHz and 0.975 at 3.5 MHz, where the loss is too large for the first-order scaling to be more than indicative. The time step alone accounts for a difference of that size; angular units or noise injected before the filter would change it further. The σ_{noise} axes of the two studies should not be identified, and no number in this paper at a given σ_{noise} should be compared with a number at the same σ_{noise} in Ref. [4]. The conclusions above do not depend on the scale: Eq. (21) holds at every noise strength, and the matched comparison is negative over the whole range.

The same applies to the training noise. Under the same assumptions, $\sigma_{\text{train}} = 1$ MHz at 2 ns corresponds to about 4 MHz at 0.1 ns, above the range evaluated in Ref. [4]. We did not train at the matched weaker level; by Eq. (21) the pulse-independence does not depend on it, but the quality of the trained agent may.

The step matters in a second way. The part of the leakage bound that responds to white noise (Sec. VD) contains a fourth finite difference of the controls and scales as Δt^{-4} : replayed on finer grids, its growth of 3.8×10^{-3} per ns at 2 ns becomes about 15, 250 and 4200 times larger at 1, 0.5 and 0.25 ns. On a grid of a thousand

steps a bound accumulated from the noisy Hamiltonian would exceed every other term of the objective by orders of magnitude, so Ref. [4] presumably evaluates it otherwise; it does not say how. The second observation of Sec. VD is therefore a property of our implementation, and we cannot tell whether it applies to theirs. The direct-optimizer test of Sec. VE and the matched comparison of TRPO agents (Table IV) were both repeated without it, and their outcome did not change.

D. Why the discrepancies arise

Some differences are simply what independent reimplementations means. The two studies differ in discretization, filtering pipeline, stopping rules, curriculum schedule, training budget and evaluation protocol, and the optimization problem is high-dimensional and non-convex; exact numerical coincidence is not the appropriate standard. The parameter studies of Appendix D make the same point from the other side: performance depends measurably on the architecture and interpretably on the UFO weights, and a single run of any configuration carries a seed-to-seed spread of several 10^{-3} in $\bar{F}$ at 1 MHz.

The main discrepancy, however, is not an implementation detail. Two properties of the UFO formulation under white control noise — the pulse-independence of Eq. (21), and the fact that a policy observing the noisy propagator learns to respond to it rather than to produce a robust open-loop pulse — follow from the formulation itself, and we would expect them in any faithful implementation. They are the reason we read the robustness results of Ref. [4] as not established, rather than as not reproduced for incidental reasons. The third, the domination of the noisy objective by the leakage bound, depends on where the bound is evaluated and on the time step (Sec. VII C); it shaped the training of our first noise-optimized agent and may not carry over. The conclusion on white-noise training does not rest on it: the direct optimizer reaches it with the bound taken from the noisy and from the noise-free controls alike (Sec. VE), and so do three pairs of agents retrained with the bound taken from the noise-free controls (Sec. VC).

E. Relation to hardware

This paper compares optimizers on a common model, and none of its fidelities is a statement about a device. Five remarks place that model against current hardware.

Noise. Frequency and amplitude noise in superconducting circuits is dominated by low-frequency components — $1/f$ flux and charge noise and slow drift of the calibration [1, 25]. Of the noise models of Sec. V the quasi-static one is therefore the closer to a device, and the white model of Ref. [4], under which no training can help, the further.

Decoherence. The model is closed. Weak relaxation and pure dephasing, at rates $\Gamma_1 = 1/T_1$ and $\Gamma_\varphi = 1/T_\varphi$ on each qubit, lower the average fidelity of any two-qubit gate of duration T by $\frac{4}{5}T(\Gamma_1 + \Gamma_\varphi)$, whatever the gate [26]: like white control noise, Markovian decoherence leaves a pulse nothing to exploit but its duration. For $T_1 = T_\varphi = 100\ \mu\text{s}$, of the order of current processors ($T_1 = 68\ \mu\text{s}$ in Ref. [27]), this is 1.1×10^{-3} for the 70 ns Adam pulse and 1.9×10^{-3} for the 116 ns TRPO pulse: comparable to the nominal infidelities of Table II and ordered, again, by duration. By the same formula the white noise of the model at 1 MHz costs as much as $T_1 = T_\varphi \approx 5\ \mu\text{s}$, and the runtime weight of Eq. (8), which prices a nanosecond at $\kappa/(\chi T_{\text{norm}}) = 1.7 \times 10^{-4}$ of gate infidelity, corresponds to $T_1 = T_\varphi \approx 12\ \mu\text{s}$. Both describe devices an order of magnitude less coherent than today’s; a weight matched to current coherence would move the optimum of Eq. (8) towards longer and more accurate pulses.

Bandwidth. The 10 MHz filter is narrow by hardware standards. Sycamore processors, descendants of the gmon, drive single-qubit gates with 25 ns pulses and perform iSWAP-like gates by turning on a 20 MHz coupling for 12 ns [28]: the amplitude bound of the present model, four to six times faster than the 54–72 ns that its filter allows for the same exchange angle (Sec. VIE). The runtimes of Sec. VI are thus set by a modelling choice of Ref. [4] more than by the coupler.

Higher levels. Those fast gates are not members of the benchmark family. The coupling shifts $|11\rangle$ through $|02\rangle$ and $|20\rangle$ by $4g^2/|\eta|$, so an exchange pulse carries a conditional phase $\phi = (4/|\eta|) \int g^2 dt$; it is $\pi/6$ for the Sycamore gate [28], and $\pi/5$ in our model for a square 20 MHz pulse of the same exchange angle. $N(\alpha, \alpha, \pi/2)$ contains no such phase and local rotations cannot remove it, which leaves an otherwise perfect pulse with $1 - F = \sin^2(\phi/4)$. This is what limits the pulses optimized without noise in Sec. VE: their nominal infidelities, 6.1, 2.8 and 1.6×10^{-4} at 70, 100 and 130 ns, equal $\sin^2(\phi/4)$ to within 1%. In this model the fidelity of an exchange pulse therefore improves only as $1/T^2$, a trade-off that native gate sets avoid by making ϕ part of the gate [16] or by cancelling it through the level structure of the coupler [29].

Absolute fidelity. Two-qubit gates on tunable-coupler transmons reach a Pauli error of 3.8×10^{-3} for a continuous fSim family on gmon qubits [16], and fidelities of 99.76% and 99.87% for CZ and iSWAP gates [29], decoherence included. Closed-system fidelities of 0.982–0.997 (Table II) are not competitive pulse designs and are not offered as such. What transfers from this study is the comparison between methods and the two structural results of Sec. VIII, not the pulses.

F. Limitations

The study is simulation-based. The model is physically motivated but truncated to three levels per mode under a

rotating-wave description, and real devices add transfer-function nonlinearities, spurious modes, drift and measurement constraints that are not represented. The noise model is Gaussian and white, applied to control amplitudes and to η ; it does not cover quasi-static drift, temporally correlated noise or open-system decoherence, and Eq. (21) is specific to it. For noise with memory the pulse shape does matter. Section VE shows that a gradient optimizer trained against quasi-static noise then gains about a fifth in sensitivity, and that agents trained under white noise do not; whether a policy trained under noise with memory finds such pulses, or better ones, we did not test, and it is the most important open question this work leaves.

The agents of Table II, of Fig. 5 and of the closed-loop test were each trained with a single seed, and the first noise-optimized agent with the leakage bound accumulated from the noisy trajectory. The matched comparison was repeated with three seeds and with the bound taken from the noise-free controls (Table IV); the closed-loop test, the selected controllers and the runtime sweep were not. Three seeds settle less than they may seem to. The noise-trained plans are less robust in each of them, but by a margin that is within the spread between seeds and that their durations account for, so the data establish that stochastic training brings no gain, not that it does harm. Conclusions that rest on larger effects are established: the absence of an order-of-magnitude robustness gain, the direct optimizer’s lead of about 0.012 in $\bar{F}$ at 1 MHz (and its explanation by duration and nominal fidelity), and the runtime results. All agents were trained at a time step of 2 ns, and their training episodes almost never stopped early (Sec. VD); an implementation in which the stopping time is learned under noise could behave differently, although Eq. (21) would still bound what its pulses can gain.

The time step of 2 ns, about twenty times coarser than that of Ref. [4], was set by the computational budget: each TRPO agent already takes 9.5×10^7 simulator steps, and the same number of episodes at 0.1 ns would take twenty times more. Its consequences for the noise scale and for the leakage bound are discussed in Sec. VIIC; we did not train at a finer step.

The direct optimizer is deliberately minimal — one initialization and 400 iterations per horizon on a 10 ns grid starting at 40 ns — and is not a survey of gradient methods such as automatic-differentiation GRAPE or Krotov. The comparison is between one RL framework and one gradient optimizer on one hardware model, not a ranking of control methodologies, and it does not include model-based RL [11]. Finally, both production TRPO agents and the curriculum’s first target were trained in stages with changing stopping costs, as documented in Table I. Nor do we survey earlier comparisons between reinforcement learning and gradient methods in quantum control. That gradients do well when the model is known is not new; what this paper adds is the mechanism of Eq. (21) and the structure of the benchmark family.

G. Relation to subsequent work

Read from the present vantage point, the lasting contribution of Ref. [4] looks less like a particular optimizer and more like a demonstration that robust analog control can be posed as a learning problem with leakage-aware costs and stochastic perturbations. The literature since has refined that in two directions. Model-free and hardware-interacting schemes reduce model bias by adapting on the device itself [10], and model-based or hybrid methods address the sample inefficiency that makes pure model-free RL expensive [11]. Our results sharpen the case for both. When the model is known and the noise is white, a direct optimizer wins both the nominal and the robustness problem, simply by producing short and accurate pulses. What a noise-trained policy does learn is feedback, which is useful only where the state can be observed during the operation. Learning therefore earns its cost where the model is uncertain or the noise has structure that a pulse can exploit — the regime of closed-loop refinement against the device, where the perturbation distribution is the device’s own rather than a modelled one [30–32].

VIII. CONCLUSION

We have independently reimplemented the UFO framework for robust analog two-qubit gate design on a multilevel gmon model — second-order TSWT leakage bound, exact bandwidth filter and TRPO training in a stochastic control environment — and compared it with a direct gradient optimizer that uses the same Hamiltonian, the same objective and the same training noise.

The framework works: it produces high-fidelity analog gates, faster than a compiled implementation, and its published hyperparameters are close to optimal in our hands. Three of its headline claims, however, are not supported in our implementation. A single-start direct optimizer beats our policy-gradient controller on every nominal metric and, against control noise, at every noise strength. Training the policy under noise does not make its deployed pulses more robust than those of an identical agent trained without noise, in any of four matched pairs of agents; what it changes is the agent’s closed-loop behaviour, which open-loop pulses do not inherit. And the order-of-magnitude runtime gain sits around a target that is the identity, measured against a generic reference; against a target-specific two-CNOT synthesis the learned pulses are at most twice as fast.

Two results explain these findings and should transfer beyond this study. Against temporally uncorrelated control noise, the open-loop robustness of any pulse is fixed to leading order by its nominal fidelity and its duration, Eq. (21), with a rate that we compute from the Hamiltonian and verify on every pulse we produced; no training procedure can improve on that except by making pulses more accurate or shorter. And every member

of the benchmark subfamily $N(\alpha, \alpha, \pi/2)$ is an exchange rotation by $\alpha - \pi/2$, which makes its runtime landscape a function of the exchange angle and fixes the fair synthesis reference at two CNOTs.

The useful question for noise-aware learning in quantum control is therefore not how large its advantage is under white noise — there, to leading order, there is none to be had — but whether it finds pulse shapes that exploit noise with memory, as a gradient optimizer trained against quasi-static noise already does to a modest degree, and whether the robustness it reports is that of a deployable open-loop pulse or of a feedback policy. Benchmarks that separate those two cases, and that state the time step and units of their noise model, would make results in this area comparable.

DATA AVAILABILITY

All code and data are public. The simulation and analysis code is released under the MIT licence at <https://github.com/FrancescoMinisini/RL-for-Universal-Analog-Quantum-Control>, and the data, the figure sources and this manuscript under CC BY 4.0 at <https://github.com/FrancescoMinisini/Document>. The released data include every training run behind this paper (arguments, per-iteration logs, checkpoints and control plans), the raw and smoothed robustness curves of Fig. 5, the matched-history and closed-loop evaluations of Fig. 6, the six runs and the evaluation behind Table IV, and the gate-structure, mirror-symmetry and first-order analyses of Secs. IIF, VI and VD. Every figure and table can be regenerated from them with the analysis scripts in the code repository, whose outputs are deterministic; a table in the data repository maps each figure to its data file, script and source run.

ACKNOWLEDGMENTS

I thank Prof. Morten Hjorth-Jensen and Prof. Davide Emilio Galli for their supervision and for many discussions that shaped this work. This study was carried out as part of a Bachelor’s thesis at the Department of Physics, Università degli Studi di Milano. AI-based tools were used as auxiliary instruments for code drafting and debugging, literature search, and stylistic revision; all derivations, modeling assumptions, numerical methodology, experimental design and results were independently verified by the author.

Appendix A: The TSWT coherent leakage bound

We summarize the derivation of Eq. (6). Starting from the decomposition Eq. (4) in the regime $\epsilon/\Delta \ll 1$, one seeks an anti-Hermitian generator $\hat{S}(t) = \epsilon\hat{S}_1 + \epsilon^2\hat{S}_2 + \dots$

such that the transformed Hamiltonian Eq. (5) is block-diagonal to the desired order. Order by order, the block-off-diagonal components must satisfy

$$[\hat{H}_0, \hat{S}_1] = -\hat{H}_2, \quad [\hat{H}_0, \hat{S}_2] = -([\hat{H}_1, \hat{S}_1] - i\dot{\hat{S}}_1)_{\text{od}}, \quad (\text{A1})$$

which are solved elementwise on the off-diagonal blocks by dividing by the subspace energy differences $E_\alpha - E_{\alpha'}$; the assumption $\epsilon/\Delta \ll 1$ is what makes these divisions safe. After second order the residual block-off-diagonal Hamiltonian is

$$H_{\text{od}}^{(3)} = [\hat{H}_1, \hat{S}_2] + \frac{1}{3} [[\hat{H}_2, \hat{S}_1], \hat{S}_1] - i\dot{\hat{S}}_2 = O\left(\frac{\epsilon^3}{\Delta^2}\right). \quad (\text{A2})$$

The leakage amplitude out of Ω_0 over $[0, T]$ is then an interaction-picture integral of $\hat{H}_{\text{od}}$ against a rapidly oscillating phase $e^{i \int \Delta dt}$. Repeated integration by parts converts this into boundary terms, which pick up $\|\hat{H}_{\text{od}}\|/\Delta$ at $t = 0$ and $t = T$, plus a residual integral controlled by the second derivative of $\hat{H}_{\text{od}}$ divided by Δ^2 . Collecting these gives Eq. (6).

The two contributions dominate in complementary regimes: for slow, off-resonant modulation the boundary terms dominate, and for faster on-resonant modulation the derivative term does. This is why the bound is usable as a single penalty across the operating range rather than only in one limit.

Appendix B: Algorithmic details

Environment step. Given the current observation and propagator: map the action to physical controls; apply the bandwidth filter Eq. (15); if stochastic training is enabled, sample Gaussian perturbations for $g, \delta_{1,2}, f_{1,2}$ and η and wrap the phases modulo 2π ; build H_t from Eq. (1); propagate $U_{t+1} = e^{-iH_t \Delta t} U_t$ by exact diagonalization; update the leakage estimator with the same, possibly noisy, H_t ; evaluate the running cost Eq. (8) and reward $r = -C$; test the stopping condition (time limit or cost threshold); emit the next observation Eq. (16).

Incremental leakage estimator. At each step: decompose H_t into H_0, H_1, H_2 ; solve Eq. (A1) for S_1 ; estimate $\dot{S}_1$ from the stored previous value by finite differences; solve for S_2 ; estimate $\dot{S}_2$; form $H_{\text{od}}^{(3)}$ from Eq. (A2) and symmetrize; take its spectral norm. The first and last samples update the boundary contribution; the discrete second derivative, formed from the two previous stored values, updates the integral contribution. Storing only the two previous states keeps the estimator $O(1)$ per step.

TRPO loop. Per iteration: collect a batch of stochastic rollouts in parallel workers; compute discounted returns and generalized advantage estimates [33]; take the KL-constrained policy step Eq. (10) by conjugate gradient with backtracking line search; refit the value network by regression on the empirical returns; evaluate the deterministic policy in the nominal environment and store

the control plan if it improves on the best so far. Checkpoints and induced control plans are saved every iteration, so that comparisons can be made at the level of the realized pulse rather than only at the level of network weights.

Adam baseline. For each candidate horizon: discretize the trajectory into $N_T = T/\Delta t$ steps; initialize all raw parameters to zero, which after the tanh map gives zero amplitudes and phases π ; for 400 steps, filter the controls, simulate one trajectory perturbed at σ_{train} , evaluate the UFO cost including the leakage bound on that noisy trajectory, back-propagate, clip the gradient norm at 10 and take an Adam step; keep the iterate with the lowest sampled cost. The optimizer’s own choice of horizon is the one with the lowest sampled cost. Each horizon’s stored plan is also re-evaluated deterministically for cost, fidelity, leakage, boundary penalty and runtime, and Table II reports both selections, the 60 ns plan chosen by the sampled cost and the 70 ns plan chosen by the deterministic one.

Appendix C: First-order response to control noise

Let the noise on channel c take the value $\xi_{c,k}$ (in MHz) during step k , so that it adds $s \xi_{c,k} \hat{G}_c$ to the Hamiltonian $\hat{H}_k$ of that step, with $s = 2\pi \times 10^{-3}$ and $\hat{G}_c$ the operator that the channel multiplies in Eq. (1). To first order the noisy propagator is $U = U_0 [1 - is \sum_{c,k} \xi_{c,k} \hat{G}_{c,k}]$, with the toggling-frame generators

$$\tilde{G}_{c,k} = W_k^\dagger \left[\int_0^{\Delta t} e^{i\hat{H}_k \tau} \hat{G}_c e^{-i\hat{H}_k \tau} d\tau \right] W_k, \quad (\text{C1})$$

where U_0 is the nominal propagator and W_k its part up to the start of step k . For zero-mean Gaussian noise with covariance $\langle \xi_{c,k} \xi_{c',l} \rangle = \delta_{cc'} C(k, l)$, and a nominal gate that reaches its target, the average fidelity (17) is, to leading order,

$$1 - \bar{F} \simeq 1 - \bar{F}(0) + \frac{d}{d+1} s^2 \sum_c \sum_{k,l} C(k, l) v_c(k, l),$$

$$v_c(k, l) = \frac{\text{Re Tr}(P \tilde{G}_{c,k} \tilde{G}_{c,l} P)}{d} - \frac{\text{Tr}(P \tilde{G}_{c,k} P) \text{Tr}(P \tilde{G}_{c,l} P)}{d^2}. \quad (\text{C2})$$

For stationary noise the double sum is the overlap of the noise spectrum with the filter function $\Phi_c(f) = \sum_{k,l} v_c(k, l) \cos[2\pi f(t_k - t_l)]$ of the pulse [20–22]. White noise, $C(k, l) = \sigma_{\text{noise}}^2 \delta_{kl}$, keeps only the diagonal $v_c(k, k)$, which is the variance V of Sec. VD evaluated on $\tilde{G}_{c,k}$ and does not depend on W_k as long as the nominal evolution keeps the computational subspace invariant: this is Eq. (21), and in the frequency domain it is the statement that the average of Φ_c over the Nyquist band is the same for every pulse of a given duration. Quasi-static noise, $C(k, l) = \sigma_{\text{noise}}^2$, keeps the whole sum, $\Phi_c(0)$,

TABLE VII. White-noise infidelity rate per channel, in $10^{-5} \text{ ns}^{-1} \text{ MHz}^{-2}$: first-order value of Eq. (C2) and Monte Carlo replay with the noise on that channel alone ($\sigma_{\text{noise}} = 1 \text{ MHz}$, 4000 realizations). Ranges are over the five pulses of Sec. V.

Channel	First order	Monte Carlo
Detuning δ_1 or δ_2	1.579–1.580	1.50–1.59
Drive amplitude f_1 or f_2	9.927–9.934	9.66–10.04
Coupling g	6.772–6.775	6.47–7.03
Anharmonicity η	< 0.002	< 0.002
All channels	29.79–29.80	28.8–29.3

TABLE VIII. Architecture sweep on $N(2.2, 2.2, \pi/2)$, averaged over three seeds. Best value in each column in bold.

Hidden sizes	$\bar{C}_{\text{eval}}$	$\bar{F}_{\text{eval}}$	$\bar{F}_{\sigma=1}$	$\bar{T}$ (ns)	$\bar{L}$ (10^{-4})
(64, 64, 64)	0.3584	0.9941	0.9637	106.7	4.05
(64, 32, 32)	0.3592	0.9950	0.9622	111.3	3.67
(64, 64, 64, 32)	0.3626	0.9926	0.9604	107.3	4.17
(128, 64, 32)	0.3630	0.9920	0.9616	106.7	4.22
(128, 64, 32, 32)	0.3691	0.9906	0.9589	110.7	3.56

in which generators at different times can add or cancel; that is where the pulse shape enters.

We evaluate Eq. (C1) exactly, in the eigenbasis of each $\hat{H}_k$, and check Eq. (C2) against Monte Carlo replays of the full dynamics with noise of the same covariance. Table VII gives the white-noise case channel by channel.

Appendix D: Fixing the free parameters

Two ingredients of the framework are not determined by the physics: the capacity of the actor-critic networks and the relative weights in Eq. (8). Both could silently drive the comparisons that follow, so we fix them first, on the representative target $N(2.2, 2.2, \pi/2)$, under a reduced budget and before any production run. Every sweep run is noise-optimized at $\sigma_{\text{train}} = 1 \text{ MHz}$.

1. Actor-critic architecture

We swept five hidden-layer layouts of comparable scale, three seeds each, at 20 policy iterations and 5000 episodes per batch (Table VIII). Selection was not made on total cost alone, because a network can reach a favourable cost by trading leakage against fidelity or by settling on an unnecessarily long pulse; we therefore tracked nominal fidelity, leakage, runtime and robustness at $\sigma_{\text{noise}} = 1 \text{ MHz}$ simultaneously.

Two conclusions follow. First, architecture is not neutral: moderate changes in width and depth produce measurable differences in the final trade-off. Second, the dependence is not monotonic. The deepest model tested achieves the *lowest* mean leakage of the sweep and simul-

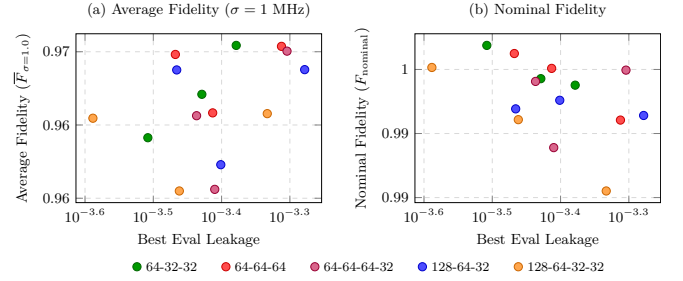


FIG. 10. Architecture trade-offs: evaluation leakage against (a) average fidelity at $\sigma_{\text{noise}} = 1 \text{ MHz}$ and (b) nominal fidelity. Leakage minimization is not a proxy for robustness — the layout with the lowest leakage, (128, 64, 32, 32), has the worst aggregate cost and the weakest robustness.

taneously the *worst* aggregate cost, the weakest robustness and the lowest nominal fidelity (Fig. 10). Minimizing one term of a multi-objective cost is not a selection principle.

The best aggregate performer is (64, 64, 64), but its margin over the paper layout (64, 32, 32) is 0.2% in evaluation cost, and (64, 32, 32) is in fact better in mean nominal fidelity and mean leakage. That margin is well inside the seed-to-seed spread: the three seeds of (64, 32, 32) alone reach $\bar{F}_{\sigma=1} = 0.959, 0.962$ and 0.965 . Given a margin that small, and a sweep run under a deliberately reduced budget — which makes this a comparison of practical trainability rather than of asymptotic capacity — we retain the paper architecture (64, 32, 32) for all production runs. This maximizes comparability with Ref. [4] at negligible empirical cost. The same spread sets the resolution of every single-seed comparison later in the paper: differences of a few 10^{-3} in $\bar{F}$ at 1 MHz between two individual runs are not significant on their own.

2. Sensitivity to the UFO weights

The weights in Eq. (8) are part of the physical definition of the problem, not optimizer settings, so they deserve separate treatment. We swept $\chi \in \{5, 10, 20\}$, $\mu \in \{0.1, 0.2, 0.4\}$, $\kappa \in \{0.05, 0.1, 0.2, 0.4\}$ at fixed $\beta = 10$ (36 configurations, two seeds each, five iterations of 10 000 episodes with stopping cost 0.15). Holding β fixed makes the sweep a study of the ratios χ/β , μ/β , κ/β , which is the physically meaningful reading and keeps the design tractable.

A methodological caveat governs the whole sweep: the best evaluation cost is *not* a valid selection criterion across configurations, because different weights define different scalarizations and their numerical costs are not commensurable. Since all runs here terminate at the same evaluation time of 180 ns, increasing κ mostly adds a rigid offset. We therefore compare on physical quantities only — nominal fidelity, leakage, and robustness at

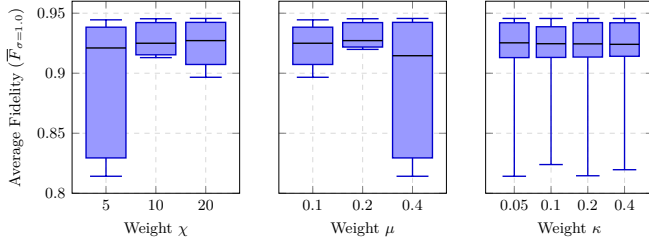


FIG. 11. Sensitivity of the average fidelity at $\sigma_{\text{noise}} = 1$ MHz to the UFO weights, shown separately for the fidelity weight χ , the boundary weight μ and the runtime weight κ . Underweighting fidelity ($\chi = 5$) opens a failure mode near $\bar{F} \approx 0.82$; $\mu = 0.4$ is strongly bimodal; κ has almost no effect in this fixed-horizon regime.

$\sigma_{\text{noise}} = 1$ MHz.

The fidelity weight has the clearest effect (Fig. 11). At $\chi = 5$ the distribution is broad and several runs collapse to $\bar{F} \approx 0.81$ – 0.83 ; at $\chi = 10$ and $\chi = 20$ it tightens around 0.92 – 0.95 . Underweighting logical accuracy rela-

tive to leakage lets the optimizer settle for controls that are clean but simply not the right gate. The boundary weight behaves non-monotonically: $\mu = 0.2$ gives the best median trade-off, $\mu = 0.1$ leaves slightly more leakage, and $\mu = 0.4$ becomes strongly bimodal, producing both some of the best runs of the sweep and its worst lower-tail failures. Larger μ does reduce the leakage proxy on average — the lowest-leakage region is often at $\mu = 0.4$ — but this does not translate into better robust control, which is the same lesson the architecture sweep taught. The runtime weight κ is nearly inert here, for the structural reason noted above; its influence reappears in Sec. VI, where gate duration is the observable rather than a fixed boundary condition.

The Pareto-favourable region is occupied jointly by $(\chi, \mu) \approx (10, 0.2)$ and $(\chi, \mu) \approx (20, 0.4)$, with the latter marginally ahead. Given two seeds and a small budget, that margin does not justify departing from the reference scalarization, so we retain the weights of Eq. (19) throughout. The useful conclusion is not the choice itself but its two components: the weights matter, in a physically interpretable way, and the original values sit close to the best region in an independent implementation.

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